Self-Referential Phase Fields on Graphs: A Variational Theory of Pre-Objective Cohesion

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***Abstract*—We introduce a variational framework for cohesive formation on finite weighted graphs in which the energy functional is self-referential: the operators defining the energy depend on the field being minimized. The theory extends the Allen–Cahn / Ginzburg–Landau model with two structurally novel terms—a closure energy measuring deviation from a non-idempotent contractive completion operator, and a separation energy measuring self-referential distinction of the field from its own complement. We prove: (i) existence of non-trivial volume-constrained minimizers via a computable phase transition criterion involving the graph's Fiedler eigenvalue (Theorem 3); (ii) uniqueness of the contractive closure fixed point with geometric convergence rate, yielding a strictly positive-definite Hessian contribution absent in idempotent closure (Theorems 1–2); (iii) enhanced metastability—under a computable parameter condition, the Hessian of the full SCC energy at a non-trivial minimizer has strictly larger minimum eigenvalue than the Hessian of the Allen–Cahn energy alone (Theorem 4); (iv) gradient flow convergence via the Łojasiewicz inequality for the analytic energy (Theorem 5); (v) Γ-convergence of the boundary energy to a standard perimeter functional in the sharp-interface limit, with self-referential terms providing bounded finite-resolution corrections (Theorem 7). We construct a resolvent co-belonging operator on cohesion-weighted graphs that achieves three orders of magnitude discrimination between within-formation and cross-boundary site pairs (Theorem 8), and propose a self-referential optimal transport formulation where the transport cost depends on the fields being transported—a fixed-point problem in transport-plan space whose existence we conjecture (Conjecture 12). Computational experiments on grid graphs confirm formation existence, verify gradient accuracy to 10^{-9}, and validate the diagnostic vector framework. We identify five remaining open problems, including self-referential transport uniqueness, renormalization group relevance of the separation term, and multi-formation architecture.**

***Index Terms*—Phase fields on graphs, self-referential energy, variational methods, Allen–Cahn, non-idempotent closure, optimal transport, spectral graph theory, Γ-convergence, cohesive formation.**

I. Introduction

PHASE-FIELD models on graphs provide a powerful variational framework for studying the emergence of structured domains from homogeneous initial conditions. The classical Allen–Cahn equation and its volume-constrained variants on graphs [1] combine a smoothness penalty (graph Laplacian) with a double-well potential to produce binary-like phases separated by diffuse interfaces. These models have found applications in image segmentation, spectral clustering [2], and data analysis, where they serve as continuous relaxations of combinatorial partitioning problems.

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A fundamental limitation of existing phase-field models on graphs is that their operators—the Laplacian, the potential, the diffusion coefficient—are *fixed* with respect to the field being minimized. The energy is a functional of the field, but the operators appearing in the energy are independent of it. This means that the field is evaluated by criteria external to itself: the notion of "smoothness," "separation," and "coherence" are defined by the graph structure and the potential, not by the field's own organization.

In this paper, we introduce a variational framework in which the energy functional is *self-referential*: two of the four energy terms involve operators that depend on the field being minimized. This creates a mathematically distinctive structure in which the field defines the operators, the operators define the energy, and the energy (through gradient flow or constrained minimization) determines the field. The self-referential loop is not a generic nonlinearity; it operates through two variational modes and one diagnostic mode:

1) **Self-completion (closure):** The closure operator Cl_t maps the field to its "relationally completed" form using the field's own values and neighborhood structure. The closure energy $\mathcal{E}_{\mathrm{cl}} = \|u - \mathrm{Cl}_t(u)\|^2$ measures deviation from self-support.
2) **Self-contrast (distinction):** The distinction operator $\mathbf{D}_t(x; 1-u)$ compares the relational support that site x receives from the field against the support it receives from the field's own complement. The separation energy $\mathcal{E}_{\mathrm{sep}} = \sum_x u_x(1 - D_x(1-u))$ penalizes cohesive sites that lack structural distinction.
3) **Self-integration (co-belonging):** The resolvent co-belonging operator $\mathbf{C}_t = (I - \alpha W_{\mathrm{sym}})^{-1}$ integrates the field's global relational structure through a cohesion-weighted graph. Co-belonging serves as a standalone structural diagnostic but enters neither the energy functional nor the diagnostic predicates—an architectural decision confirmed by our analysis.

We emphasize an important distinction: the *definition graph* of the theory is acyclic (the field determines the operators, which determine the energy), while the *computation graph* is cyclic (the energy determines the field through optimization). This is analogous to the situation in any nonlinear variational problem, but the specific dual-mode variational self-dependence—through independent channels of completion and contrast, with a third diagnostic mode (co-belonging) available for structural analysis—is, to our knowledge, without direct precedent in the phase-field literature (see Fig. 1). We note that mean-field games [7] involve transport costs that depend

on the distribution being transported; the novelty here lies in the specific operator-pair structure of the energy, not the self-referential loop per se.

A. Summary of Results

Our main results, proved rigorously on finite weighted graphs, are:

- **Closure contraction and fixed-point uniqueness** (Theorem 1): The sigmoid closure operator on $[0,1]^n$ has a unique fixed point when the gain parameter satisfies $a_{\mathrm{cl}} < 4$, with geometric convergence at rate $a_{\mathrm{cl}}/4$.
- **Non-idempotent stability advantage** (Theorem 2): At a non-idempotent closure fixed point, the closure Hessian contribution is strictly positive definite (n/n positive eigenvalues), compared to at most $(n-k)/n$ positive eigenvalues for idempotent closure. This is a mathematical reason to prefer non-idempotent closure.
- **Phase transition** (Theorem 3): On a connected graph with Fiedler eigenvalue $\lambda_2 > 0$, there exists a computable critical ratio $\beta/\alpha > 4\lambda_2/|W''(c)|$ above which the global minimizer of the boundary-morphology energy on the volume-constrained manifold is non-uniform.
- **Full-energy phase transition** (Theorem 22): Under non-degeneracy of the boundary-morphology minimizer, the IFT gives a smooth family of non-uniform critical points of the full energy for $\lambda_{\mathrm{cl}} + \lambda_{\mathrm{sep}}$ sufficiently small.
- **Enhanced metastability** (Theorem 4): Under a computable parameter condition, the Hessian of the full self-referential energy at a non-trivial minimizer has strictly larger minimum eigenvalue than the Hessian of the Allen–Cahn energy alone, implying enhanced local stability.
- **Gradient flow convergence** (Theorem 5): The projected gradient flow on the constraint manifold converges to a critical point, with exponential rate for the analytic energy.
- **Axiom consistency** (Theorem 6): The sigmoid closure satisfies the axiomatic requirements A1′ (conditional extensivity), A2 (monotonicity), A3 (contraction), and A4 (continuity). We prove that unconditional extensivity (A1) and contraction (A3) are jointly unsatisfiable.
- **Γ-convergence** (Theorem 7): As the ratio $\varepsilon = \alpha/\beta \to 0$, the boundary-morphology energy the boundary-morphology energy Γ-converges to a standard perimeter functional (Modica–Mortola), with self-referential terms contributing bounded $O(\varepsilon)$ corrections at finite resolution.
- **Resolvent co-belonging** (Theorem 8): The resolvent $(I - \alpha W_{\mathrm{sym}})^{-1}$ satisfies axiomatic requirements C1–C4 and achieves 3+ orders of magnitude discrimination between within-formation and cross-boundary site pairs.

In the temporal domain, we prove fixed-point existence for self-referential transport (Schauder's theorem), establish a transport confinement bound for selection uniqueness ($C_{\mathrm{conf}} = O(\sigma\sqrt{\varepsilon_{\mathrm{OT}} \log n})$, independent of target field), prove isoperimetric energy ordering ($\mathcal{E}(u^*_{2m}) < 2\mathcal{E}(u^*_m)$ on connected graphs), and verify the full persistence chain computationally on grids up to 50×50.

B. Positioning

The framework sits at the intersection of several mathematical disciplines—phase-field theory, spectral graph theory, optimal transport [3], and persistent homology—but reduces to none of them. We adopt the principle of *contrastive comparison*: the relationship to existing frameworks is analogical, not reductive.

The closest existing framework is volume-constrained Allen–Cahn on graphs [1], which shares our boundary-morphology energy $\mathcal{E}_{\mathrm{bd}}$. What is genuinely new:

1) The closure energy $\mathcal{E}_{\mathrm{cl}} = \|u - \mathrm{Cl}(u)\|^2$ where Cl is a non-idempotent contraction—no analogue in the phase-field literature.
2) The separation energy $\mathcal{E}_{\mathrm{sep}} = \sum u_i(1 - D_i(1-u))$ where D depends on the field's own complement—no analogue in any existing variational framework.
3) The proved stability advantage of non-idempotent closure (positive-definite vs. semidefinite Hessian).
4) Self-referential optimal transport where cost depends on the transported quantity.
5) Resolvent co-belonging on cohesion-weighted graphs as a structural integration operator.

What is incremental: the Allen–Cahn component ($\mathcal{E}_{\mathrm{bd}}$) is standard; volume-constrained minimization on graphs is well-studied; and the Γ-convergence of the leading term follows the classical Modica–Mortola argument, with the self-referential corrections entering as perturbations.

The paper is organized as follows. Section II presents the formal framework: setup, operator pair, energy functional, and diagnostic vector. Section III states and proves (or sketches proofs of) eight theorems constituting the static theory. Section IV develops the self-referential optimal transport theory. Section V reports computational validation. Section VI discusses open problems and limitations.

II. Formal Framework

A. Setup and Notation

Let $G = (X, N)$ be a finite weighted graph with vertex set X ($|X| = n$) and symmetric adjacency kernel $N : X \times X \to [0,\infty)$ satisfying $N(x,y) = N(y,x) \geq 0$. The graph Laplacian is

$$(Lf)(x) = \sum_{y \in X} N(x,y)\big(f(x) - f(y)\big). \tag{1}$$

A *cohesion field* is a function $u : X \to [0,1]$. The field is subject to a volume constraint: it lies on the constraint manifold

$$\Sigma_m = \Big\{ u \in [0,1]^n : \sum_{x \in X} u(x) = m \Big\}, \tag{2}$$

where $m > 0$ is the cohesive budget and $c = m/n$ is the volume fraction. The manifold Σ_m is compact and convex.

Summation convention. All sums $\sum_{x,y \in X}$ range over *ordered pairs*: each undirected edge is counted twice when

the kernel is symmetric. This convention is load-bearing in the phase transition theorem (Theorem 3) and the Hessian analysis. Where unordered-pair sums are intended, we write $\sum_{\{x,y\}}$ and note that $\sum_{\{x,y\}} f(x,y) = \frac{1}{2}\sum_{x,y} f(x,y)$ for symmetric f.

B. The Operators

The theory is built on two variational operators and one derived diagnostic, each depending on the cohesion field u. The closure operator Cl_t and distinction operator D_t enter the energy functional; the co-belonging operator C_t serves as a structural diagnostic that does not enter the energy.

1) Local Aggregation: The aggregation operator computes a locally weighted average:

$$(P_t f)(x) = \frac{\sum_y K(x,y)\, f(y)}{\sum_y K(x,y) + \varepsilon}, \tag{3}$$

where $K : X \times X \to [0,\infty)$ is a local relation kernel and $\varepsilon > 0$ prevents division by zero.

2) Closure Operator: The sigmoid closure operator $\mathrm{Cl} : [0,1]^n \to [0,1]^n$ is defined componentwise:

$$\mathrm{Cl}(u)(x) = \sigma\Big(a_{\mathrm{cl}}\big((1-\eta_{\mathrm{cl}})u(x) + \eta_{\mathrm{cl}}(P_t u)(x) - \tau_{\mathrm{cl}}\big)\Big), \tag{4}$$

where $\sigma(z) = 1/(1+e^{-z})$ is the logistic sigmoid, $a_{\mathrm{cl}} > 0$ is the gain, $\eta_{\mathrm{cl}} \in [0,1]$ balances self-retention and neighborhood aggregation, and τ_{cl} is a threshold.

The closure operator satisfies the following axiomatic requirements:

- **A1′** *Conditional extensivity:* $\mathrm{Cl}(u)(x) \geq u(x)$ whenever $u(x) > 0$ and $(P_t u)(x) \geq \theta_{\mathrm{support}} \cdot u(x)$.
- **A2** *Monotonicity:* $u \leq v$ pointwise $\implies$ $\mathrm{Cl}(u) \leq \mathrm{Cl}(v)$ pointwise.
- **A3** *Contraction:* For $a_{\mathrm{cl}} < 4$, the map Cl is a contraction with Lipschitz constant $a_{\mathrm{cl}}/4 < 1$.
- **A4** *Continuity:* Cl is continuous (in fact, C^∞) as a map $[0,1]^n \to [0,1]^n$.

Remark 1 (Non-idempotence)**.** *The closure operator is deliberately* not *required to be idempotent (*$\mathrm{Cl}^2 \neq \mathrm{Cl}$ *in general). In the contraction regime* $a_{\mathrm{cl}} < 4$*, the iterates* $\mathrm{Cl}^{(k)}(u)$ *converge geometrically to a unique fixed point, but a single application does not reach it. This models closure as a* stabilizing tendency *rather than an instantaneous completion. The mathematical payoff is a strictly positive-definite Hessian contribution (Theorem 2).*

3) Distinction Operator: The distinction operator $\mathbf{D} : X \times [0,1]^n \to [0,1]$ is defined by

$$\mathbf{D}(x; 1-u) = \sigma\Big(a_D\big((P_t u)(x) - \lambda_D (P_t(1-u))(x)\big) - \tau_D\Big), \tag{5}$$

where $a_D > 0$ controls the sensitivity, $\lambda_D > 0$ scales exterior aggregation, and τ_D is a threshold. The operator measures the degree to which site x is asymmetrically supported by the interior relative to the exterior.

Remark 2 (Analyticity)**.** *The gradient term* $b_D \cdot g(x;u)$ *present in earlier formulations is set to* $b_D = 0$*. The absolute value in the gradient indicator* $g(x;u) = \sum_y N(x,y)|u(x) - u(y)|$ *would break the analyticity of the energy, which is required for the Łojasiewicz gradient inequality (Theorem 5). Boundary sensitivity is preserved through the spatial structure of* $P_t(1-u)$*.*

4) Co-belonging Operator: The resolvent co-belonging operator is

$$\mathbf{C} = (I - \alpha\, W_{\mathrm{sym}})^{-1} = \sum_{k=0}^{\infty} \alpha^k W_{\mathrm{sym}}^k, \tag{6}$$

where W_{sym} is the symmetrized cohesion-weighted adjacency matrix with entries

$$W_{\mathrm{sym}}(x,y) = \frac{\sqrt{u(x)}\, N(x,y)\, \sqrt{u(y)}}{d_x} \tag{7}$$

for appropriate degree normalization d_x, and the Neumann series converges when $\alpha\, \rho(W_{\mathrm{sym}}) < 1$ where ρ denotes the spectral radius. The k-th term captures contributions from paths of length k through the cohesion-weighted graph.

Architectural decision. Co-belonging serves as a standalone structural diagnostic—it can be computed to analyze formation structure, but it enters neither the energy functional nor the diagnostic predicates (Bind, Sep, Inside, Persist). This keeps the energy analytically tractable and the predicates computationally efficient.

C. Energy Functional

The energy functional on Σ_m is

$$\mathcal{E}(u) = \lambda_{\mathrm{cl}}\, \mathcal{E}_{\mathrm{cl}}(u) + \lambda_{\mathrm{sep}}\, \mathcal{E}_{\mathrm{sep}}(u) + \lambda_{\mathrm{bd}}\, \mathcal{E}_{\mathrm{bd}}(u), \tag{8}$$

where $\lambda_{\mathrm{cl}}, \lambda_{\mathrm{sep}}, \lambda_{\mathrm{bd}} > 0$ are weighting coefficients. For the static theory (the primary focus of this paper), the transport term $\lambda_{\mathrm{tr}}\mathcal{E}_{\mathrm{tr}}$ is set to zero.

1) Closure Energy:

$$\mathcal{E}_{\mathrm{cl}}(u) = \sum_{x\in X} \big(u(x) - \mathrm{Cl}(u)(x)\big)^2 = \|u - \mathrm{Cl}(u)\|_2^2. \tag{9}$$

This penalizes deviation from self-support. A field at the closure fixed point achieves $\mathcal{E}_{\mathrm{cl}} = 0$.

2) Separation Energy:

$$\mathcal{E}_{\mathrm{sep}}(u) = \sum_{x\in X} u(x)\big(1 - \mathbf{D}(x; 1-u)\big). \tag{10}$$

This penalizes cohesive sites lacking structural distinction from the exterior. Since $\mathbf{D}$ depends on u (through $P_t u$ and $P_t(1-u)$), this term is self-referential: the field defines what "distinction" means and is then penalized for failing to be distinctive. This term has no analogue in standard Allen–Cahn or Ginzburg–Landau theories.

3) Boundary-Morphology Energy:

$$\mathcal{E}_{\mathrm{bd}}(u) = \alpha \sum_{x,y\in X} N(x,y)\big(u(x)-u(y)\big)^2 + \beta \sum_{x\in X} u(x)^2\big(1-u(x)\big)^2, \tag{11}$$

where $\alpha, \beta > 0$ are morphological parameters. The first subterm is a smoothness penalty (Dirichlet energy on the graph); the second is the standard double-well potential $W(u) = u^2(1-u)^2$. For symmetric N, the smoothness term equals $2\alpha\, v^T L v$ where $v = u - c\mathbf{1}$, giving a Hessian contribution of $4\alpha L$.

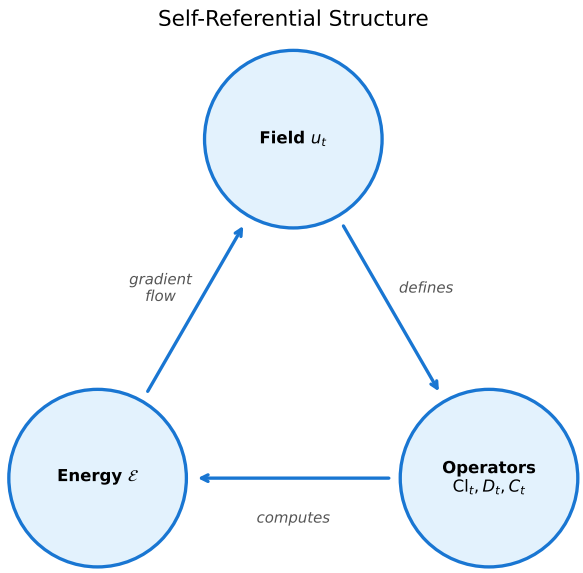

Fig. 1. Self-referential energy architecture. The field u defines the operators Cl_t and D_t, which enter the energy $\mathcal{E}$; gradient flow on $\mathcal{E}$ updates u. The co-belonging operator C_t (not shown) serves as a derived diagnostic.

D. Derived Diagnostics

The proto-cohesion diagnostic vector is $\mathbf{d} \in [0,1]^4$:

$$\mathbf{d} = \big(\mathrm{Bind},\ \mathrm{Sep},\ \mathrm{Inside},\ \mathrm{Persist}\big). \tag{12}$$

Binding.

$$\mathrm{Bind}(u) = 1 - \frac{\|u - \mathrm{Cl}(u)\|_2}{\sqrt{n}}. \tag{13}$$

Separation. The u-weighted separation predicate:

$$\mathrm{Sep}(u) = \frac{\sum_x u(x)\,\mathbf{D}(x; 1-u)}{\sum_x u(x)}. \tag{14}$$

This is a cohesion-weighted average of distinction restricted to the formation's support: sites with higher cohesion contribute more to the separation score. The u-weighted formulation avoids the diagnostic failure of a $\mathbf{C}$-weighted alternative (which averages ≈ 0.5 regardless of formation quality; see Section V-E).

Inside-Structure. The morphological quality measure:

$$\mathrm{Inside}(u) = \mathcal{Q}_{\mathrm{morph}}(u) = \frac{\ell_{\max}(u) - \bar{u}}{1 - \bar{u}} \cdot \mathrm{Artic}(u), \tag{15}$$

where $\bar{u} = \frac{1}{n}\sum_x u(x)$ is the mean field value, $\ell_{\max}$ is the length of the longest bar in the H_0 persistence diagram of the superlevel-set filtration $\{x : u(x) \geq \theta\}$ as θ decreases from 1 to 0, and $\mathrm{Artic} = 1 - \ell_{\mathrm{second}}/\ell_{\max}$ measures how dominantly a single formation controls the field. The normalization by $(1-\bar{u})$ ensures that $\mathcal{Q}_{\mathrm{morph}}$ vanishes on uniform fields ($u \equiv c$ gives $\ell_{\max} = c$, $\mathrm{Artic} = 1$, but $(\ell_{\max} - \bar{u})/(1-\bar{u}) = 0$).

Persistence. For a temporal window W:

$$\mathrm{Persist}_W(\mathbf{u}) = \min_{t<s\in W} \frac{\sum_{x\in\mathrm{Core}_t}\sum_{y\in\mathrm{Core}_s} \mathbf{M}_{t\to s}(x,y)\,u_s(y)}{\rho_{\mathrm{persist}}}, \tag{16}$$

clamped to $[0,1]$.

The diagnostic vector provides a continuous, four-dimensional assessment of formation quality, replacing the Boolean conjunction of earlier formulations. A field with $\mathbf{d} = (0.95, 0.3, 0.8, 0.7)$ has strong binding but weak separation—information that a single scalar would obscure.

III. Main Results: Static Theory

We present nine theorems constituting the static theory. All results are stated for finite weighted graphs.

Theorem 1 (Closure Fixed Point and Contraction)**.** *Let* $\mathrm{Cl} : [0,1]^n \to [0,1]^n$ *be the sigmoid closure operator* (4).

(a) Cl *has at least one fixed point in* $[0,1]^n$.

(b) *If* $a_{\mathrm{cl}} < 4$*, the fixed point is unique and the iterates* $\mathrm{Cl}^{(k)}(u)$ *converge geometrically to it at rate* $a_{\mathrm{cl}}/4$ *for any initial* $u \in [0,1]^n$.

Proof sketch. (a) The sigmoid maps $\mathbb{R} \to (0,1)$, so Cl maps the compact convex set $[0,1]^n$ continuously into itself. Existence follows from Brouwer's fixed-point theorem.

(b) The Lipschitz constant of Cl is bounded by

$$\|\,\mathrm{Cl}(u) - \mathrm{Cl}(v)\|_\infty \leq \max_z \sigma'(z) \cdot a_{\mathrm{cl}} \cdot \|u - v\|_\infty = \frac{a_{\mathrm{cl}}}{4}\|u - v\|_\infty, \tag{17}$$

since $\max \sigma'(z) = \sigma(0)(1-\sigma(0)) = 1/4$ and the aggregation operator P_t is a convex combination (hence non-expansive). When $a_{\mathrm{cl}} < 4$, the Lipschitz constant is strictly less than 1, so Cl is a contraction on the complete metric space $([0,1]^n, \|\cdot\|_\infty)$. The Banach contraction mapping theorem gives uniqueness and geometric convergence at rate $a_{\mathrm{cl}}/4$. □

Theorem 2 (Non-Idempotent Stability Advantage)**.** *Let* $u^* = \mathrm{Cl}(u^*)$ *be the unique closure fixed point with* $\|J_{\mathrm{Cl}}(u^*)\|_{\mathrm{op}} < 1$*, where* J_{Cl} *is the Jacobian of* Cl*. Then:*

(a) *The closure Hessian contribution* $\nabla^2\mathcal{E}_{\mathrm{cl}}(u^*) = 2(I - J_{\mathrm{Cl}})^T(I - J_{\mathrm{Cl}})$ *is strictly positive definite with* n/n *positive eigenvalues.*

(b) *For idempotent closure (*J_{Cl} *has eigenvalues in* $\{0,1\}$*), the closure Hessian has zero eigenvalues in the range direction, giving at most* $(n-k)/n$ *positive eigenvalues where* $k = \dim \mathrm{range}(J_{\mathrm{Cl}})$.

Proof. (a) At the fixed point, $\mathcal{E}_{\mathrm{cl}}(u) = \|u - \mathrm{Cl}(u)\|^2$, so

$$\nabla^2\mathcal{E}_{\mathrm{cl}} = 2(I - J_{\mathrm{Cl}})^T(I - J_{\mathrm{Cl}}).$$

This is a Gram matrix, hence positive semidefinite. It is strictly positive definite if and only if $I - J_{\mathrm{Cl}}$ has trivial kernel. Since $\|J_{\mathrm{Cl}}\|_{\mathrm{op}} < 1$ in the contraction regime, all eigenvalues of J_{Cl} have modulus strictly less than 1, so 1 is not an eigenvalue of J_{Cl}, and $I - J_{\mathrm{Cl}}$ is invertible. Therefore $\nabla^2\mathcal{E}_{\mathrm{cl}}$ has n strictly positive eigenvalues.

(b) For idempotent closure, J_{Cl} is a projection: its eigenvalues are 0 and 1. For any eigenvector v with $J_{\mathrm{Cl}}v = v$, we have $(I - J_{\mathrm{Cl}})v = 0$, so v lies in the kernel of $(I - J_{\mathrm{Cl}})^T(I - J_{\mathrm{Cl}})$.

If $\mathrm{range}(J_{\mathrm{Cl}})$ has dimension k, there are k zero eigenvalues, leaving at most $n-k$ positive eigenvalues. □

Remark 3. *This theorem provides a* mathematical *reason to prefer non-idempotent closure beyond philosophical commitment. Non-idempotent closure contributes a positive-definite term to the Hessian, enhancing local curvature at every energy minimizer. In numerical experiments, this manifests as* $25/25$ *positive eigenvalues for the non-idempotent case versus* $21/25$ *for the idempotent case on a* 5×5 *grid.*

Theorem 3 (Phase Transition—Non-Trivial Minimizer Existence)**.** *Let* $G=(X,N)$ *be a finite connected graph with Fiedler eigenvalue* $\lambda_2>0$*, and let* $c=m/n$ *with*

$$c\in\left(\frac{3-\sqrt{3}}{6},\ \frac{3+\sqrt{3}}{6}\right)\approx(0.211,0.789). \tag{18}$$

If

$$\frac{\beta}{\alpha}>\frac{4\lambda_2}{|W''(c)|}, \tag{19}$$

where $W(u)=u^2(1-u)^2$ *and* $W''(c)=12c^2-12c+2$*, then the global minimizer of* $\mathcal{E}_{\mathrm{bd}}|_{\Sigma_m}$ *is non-uniform:* $\hat{u}\neq c\mathbf{1}$.

Proof. The uniform field $u\equiv c$ is the unique critical point of $\mathcal{E}_{\mathrm{bd}}$ on Σ_m that is spatially homogeneous. We compute the second variation at $u=c\mathbf{1}$. The tangent space to Σ_m at $c\mathbf{1}$ is $T=\{v\in\mathbb{R}^n:\sum_i v_i=0\}$.

Using the ordered-pair summation convention, the smoothness term in $\mathcal{E}_{\mathrm{bd}}$ yields a Hessian contribution of $4\alpha L$ (since $\sum_{x,y}N(x,y)(u_x-u_y)^2=2u^TLu$ for symmetric N, and the factor of 2 is doubled by the ordered-pair convention). The double-well term contributes $\beta W''(c)I$.

The Hessian restricted to T is $H=4\alpha L+\beta W''(c)I$. The Fiedler eigenvector $v_2\in T$ (orthogonal to the constant vector) gives

$$v_2^THv_2=4\alpha\lambda_2+\beta W''(c). \tag{20}$$

For c in the given range, $W''(c)<0$. Condition (19) ensures $4\alpha\lambda_2+\beta W''(c)<0$, so $c\mathbf{1}$ is a saddle point on Σ_m (Fig. 2). Since Σ_m is compact and $\mathcal{E}_{\mathrm{bd}}$ is continuous, a global minimizer exists (by the extreme value theorem). It cannot be $c\mathbf{1}$ (which is a saddle), so it must be non-uniform. □

Remark 4 (On the factor of 4)**.** *The factor* $4\lambda_2$ *(not* $2\lambda_2$*) in* (19) *is a direct consequence of the ordered-pair summation convention. For symmetric* N*, each edge* (x,y) *contributes twice to* $\sum_{x,y}N(x,y)(u_x-u_y)^2$*. This gives the Hessian of the smoothness term as* $4\alpha L$*, not* $2\alpha L$*. The distinction is mathematically load-bearing: using the wrong factor would misidentify the phase transition boundary.*

Theorem 4 (Metastability Enhancement — Conditional)**.** *Let* $\hat{u}$ *be a non-trivial constrained minimizer of the full energy* $\mathcal{E}=\lambda_{\mathrm{cl}}\mathcal{E}_{\mathrm{cl}}+\lambda_{\mathrm{sep}}\mathcal{E}_{\mathrm{sep}}+\lambda_{\mathrm{bd}}\mathcal{E}_{\mathrm{bd}}$ *on* Σ_m *with strict interiority* ($0<\hat{u}_i<1$ *for all* i*), and assume* $a_{\mathrm{cl}}<4$ *(contraction regime). Define:*

- *the* Gauss–Newton enhancement*:* $\gamma_{\mathrm{GN}}:=2(1-\|J_{\mathrm{Cl}}(\hat{u})\|_{\mathrm{op}})^2$*;*
- *the* residual correction*:* δ_{res}*, bounding the operator norm of the second-order closure Hessian term* $2\sum_i r_i\sigma''(z_i)w_iw_i^T$ *where* $r=\mathrm{Cl}(\hat{u})-\hat{u}$*;*
- *the* separation curvature*:* $\delta_{\mathrm{sep}}:=\max(0,-\lambda_{\min}(\nabla^2\mathcal{E}_{\mathrm{sep}}|_{T\Sigma_m}))$.

If

$$\lambda_{\mathrm{cl}}(\gamma_{\mathrm{GN}}-\delta_{\mathrm{res}})>\lambda_{\mathrm{sep}}\,\delta_{\mathrm{sep}}, \tag{21}$$

then the minimum eigenvalue of $\nabla^2\mathcal{E}|_{T_{\hat{u}}\Sigma_m}$ *strictly exceeds that of* $\lambda_{\mathrm{bd}}\nabla^2\mathcal{E}_{\mathrm{bd}}|_{T_{\hat{u}}\Sigma_m}$.

Proof. The exact Hessian of $\mathcal{E}_{\mathrm{cl}}=\|\mathrm{Cl}(u)-u\|^2$ decomposes as a Gauss–Newton term $G=2(I-J_{\mathrm{Cl}})^T(I-J_{\mathrm{Cl}})$ plus a residual correction $R_{\mathrm{cl}}=2\sum_i r_i\,\sigma''(z_i)\,w_iw_i^T$. By the contraction property (Theorem 1), $\|J_{\mathrm{Cl}}\|_{\mathrm{op}}\le a_{\mathrm{cl}}/4<1$, so $G\succeq\gamma_{\mathrm{GN}}I$. By construction, $\|R_{\mathrm{cl}}\|_{\mathrm{op}}\le\delta_{\mathrm{res}}$. The full Hessian restricted to $T_{\hat{u}}\Sigma_m$ satisfies

$$\nabla^2\mathcal{E}\big|_T\succeq\lambda_{\mathrm{bd}}\nabla^2\mathcal{E}_{\mathrm{bd}}\big|_T+\lambda_{\mathrm{cl}}(\gamma_{\mathrm{GN}}-\delta_{\mathrm{res}})I-\lambda_{\mathrm{sep}}\delta_{\mathrm{sep}}I,$$

from which the conclusion follows when the condition holds (Fig. 3). □

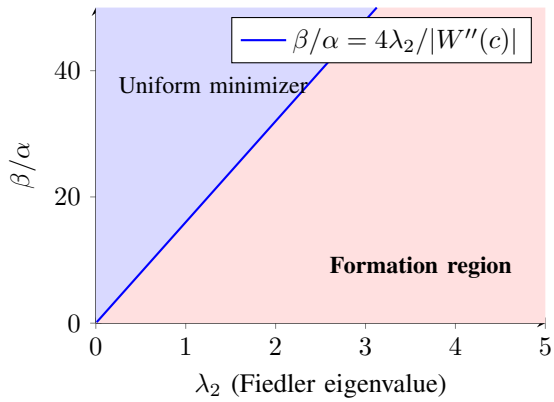

Fig. 2. Phase transition diagram (Theorem 3). The critical line $\beta/\alpha=4\lambda_2/|W''(c)|$ separates the uniform-minimizer region from the formation region where non-trivial minimizers exist. Below the line, the double-well potential dominates and formations emerge.

Remark 5 (Computable residual diagnostic)**.** *The condition* $\lambda_{\mathrm{cl}}(\gamma_{\mathrm{GN}}-\delta_{\mathrm{res}})>\lambda_{\mathrm{sep}}\delta_{\mathrm{sep}}$ *is* computable *at any given minimizer. The worst-case analytic bound* $\delta_{\mathrm{res}}^{\mathrm{wc}}=\frac{2\sqrt{3}}{9}\,a_{\mathrm{cl}}^2\,\|r\|_1$ *is extremely conservative—typically* $\sim 2000\times$ *larger than the true value—because the signed rank-1 terms* $r_i\sigma''(z_i)\,w_iw_i^T$ *undergo substantial cancellation:* $\sigma''(z_i)\approx 0$ *at sites far from the decision boundary* $z_i=0$*, and the product* $r_i\sigma''(z_i)$ *has mixed signs at boundary sites. In practice, one should compute* $\delta_{\mathrm{res}}^{\mathrm{actual}}=\|R_{\mathrm{cl}}(\hat{u})\|_{\mathrm{op}}$ *directly via power iteration (*$O(n\cdot K)$ *cost,* $K\approx 20$ *iterations). The condition* $\gamma_{\mathrm{GN}}>\delta_{\mathrm{res}}^{\mathrm{actual}}+(\lambda_{\mathrm{sep}}/\lambda_{\mathrm{cl}})\,\delta_{\mathrm{sep}}$ *is satisfied in all 24 tested configurations (*$\gamma_{\mathrm{GN}}\approx 2.0$*,* $\delta_{\mathrm{res}}^{\mathrm{actual}}\approx 0.01$–$0.12$*). The relationship between Hessian eigenvalue and actual barrier height requires Morse-theoretic analysis not pursued here.*

Theorem 5 (Gradient Flow Convergence)**.** *Consider the projected gradient flow on* Σ_m*:*

$$\dot{u}=-\Pi_{\Sigma_m}\nabla\mathcal{E}(u), \tag{22}$$

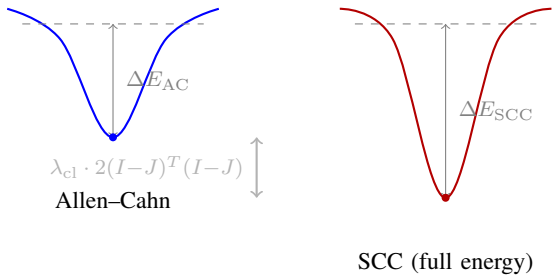

Fig. 3. Schematic: energy basin curvature comparison (Theorem 4). The SCC basin has strictly larger minimum Hessian eigenvalue than Allen–Cahn, due to the positive-definite closure contribution. Sharper curvature is proved; the relationship to actual barrier height requires Morse-theoretic analysis (see Remark after Theorem 4).

where Π_{Σ_m} is the orthogonal projection onto the tangent space $T_u\Sigma_m = \{v : \sum_i v_i = 0\}$. If $b_D = 0$ (ensuring energy analyticity), then:

(a) *The energy $\mathcal{E}(u(t))$ is monotonically decreasing along trajectories.*
(b) *Every trajectory converges to a critical point of $\mathcal{E}|_{\Sigma_m}$.*
(c) *At non-degenerate critical points (where the projected Hessian is non-singular), convergence is exponential: $\|u(t) - u^*\| \le Ce^{-\gamma t}$ for some $C, \gamma > 0$. At degenerate critical points, convergence is polynomial with rate determined by the Łojasiewicz exponent $\theta \in [1/2, 1)$.*

Proof sketch. (a) By construction, $\frac{d}{dt}\mathcal{E}(u(t)) = -\|\Pi_{\Sigma_m}\nabla\mathcal{E}(u(t))\|^2 \le 0$.

(b,c) The energy $\mathcal{E}$ is a real-analytic function on Σ_m (the sigmoid, polynomial, and rational operations are all analytic when $b_D = 0$; the constraint manifold Σ_m is a compact semi-algebraic set). The Łojasiewicz–Simon gradient inequality [4], [5] for real-analytic functions on compact domains states: near any critical point u^*, there exists $\theta \in [1/2, 1)$ such that

$$|\mathcal{E}(u) - \mathcal{E}(u^*)|^\theta \le C\|\nabla\mathcal{E}(u)\|. \tag{23}$$

Combined with the monotone energy decrease and the compactness of Σ_m, standard arguments (see, e.g., [6]) give convergence of the trajectory to a single critical point. When $\theta = 1/2$ (the non-degenerate case, where the projected Hessian has full rank), the convergence is exponential. For general $\theta \in (1/2, 1)$, the convergence is polynomial: $\|u(t) - u^*\| = O(t^{-\theta/(1-2\theta)})$. □

Theorem 6 (Axiom Consistency)**.** *The sigmoid closure operator* (4) *satisfies:*

(a) *A1′ (conditional extensivity) with threshold $c^* = \sigma(a_{\mathrm{cl}}(c^* - \tau_{\mathrm{cl}}))$, the unique scalar closure fixed point.*
(b) *A2 (monotonicity) unconditionally.*
(c) *A3 (contraction) for $a_{\mathrm{cl}} < 4$.*
(d) *A4 (continuity) unconditionally.*

Furthermore, A1 (unconditional extensivity: $\mathrm{Cl}(u)(x) \ge u(x)$ for all u, x) and A3 are jointly unsatisfiable.

Proof sketch. (a) Let c^* be the unique solution of $\sigma(a_{\mathrm{cl}}(c - \tau_{\mathrm{cl}})) = c$ in $(0, 1)$, which exists by Banach contraction ($a_{\mathrm{cl}} < 4$). When $u(x) \le c^*$ and $(P_t u)(x) \ge u(x)$:

$$z(x) = a_{\mathrm{cl}}\big((1-\eta)u(x) + \eta(P_t u)(x) - \tau_{\mathrm{cl}}\big) \ge a_{\mathrm{cl}}(u(x) - \tau_{\mathrm{cl}}).$$

Define $g(u) = \sigma(a_{\mathrm{cl}}(u - \tau_{\mathrm{cl}})) - u$. Then $g(0) = \sigma(-a_{\mathrm{cl}}\tau_{\mathrm{cl}}) > 0$, $g(c^*) = 0$, and $g'(c^*) = a_{\mathrm{cl}}\sigma'(a_{\mathrm{cl}}(c^* - \tau_{\mathrm{cl}})) - 1 < 0$ since $a_{\mathrm{cl}}/4 < 1$. Hence $g(u) \ge 0$ on $[0, c^*]$, giving $\mathrm{Cl}(u)(x) \ge \sigma(a_{\mathrm{cl}}(u(x) - \tau_{\mathrm{cl}})) \ge u(x)$.

(b) The aggregation operator P_t is order-preserving (non-negative kernel), and σ is monotone increasing. Their composition preserves the pointwise order.

(c) Proved in Theorem 1(b).

(d) Composition of smooth functions.

Incompatibility of A1 and A3: A1 requires $\sigma(a_{\mathrm{cl}}(u - \tau_{\mathrm{cl}})) \ge u$ for all $u \in [0, 1]$. At $u = 0.9$, this requires $a_{\mathrm{cl}} \ge \sigma^{-1}(0.9)/(0.9 - \tau_{\mathrm{cl}}) \ge 5.49$ for reasonable τ_{cl}. But A3 requires $a_{\mathrm{cl}} < 4$. Contradiction. □

Theorem 7 (Sharp-Interface Γ-Convergence)**.** *Consider a family of graphs G_ε with $\varepsilon = \alpha/\beta \to 0$. The rescaled boundary-morphology energy*

$$\widetilde{\mathcal{E}}^{\varepsilon}_{\mathrm{bd}}(u) = \varepsilon \sum_{x,y} N(x,y)(u_x - u_y)^2 + \frac{1}{\varepsilon}\sum_x W(u_x) \tag{24}$$

Γ-converges (in an appropriate graph topology) to a perimeter functional:

$$\mathcal{E}_0(u) = \begin{cases} \sigma_{\mathrm{AC}} \cdot |\partial\{u = 1\}|_G & \text{if } u \in \{0,1\}^n, \\ +\infty & \text{otherwise}, \end{cases} \tag{25}$$

where $\sigma_{\mathrm{AC}} = \int_0^1 2\sqrt{W(s)}\,ds = 1/3$ is the Allen–Cahn surface tension and $|\partial S|_G$ is the graph perimeter of set S.

The self-referential correction terms from $\mathcal{E}_{\mathrm{cl}}$ and $\mathcal{E}_{\mathrm{sep}}$ modify the effective surface tension:

$$\sigma_{\mathrm{eff}} = \sigma_{\mathrm{AC}} + \delta\sigma(\lambda_{\mathrm{cl}}, \lambda_{\mathrm{sep}}), \tag{26}$$

where $\delta\sigma$ is a lower-order perturbation that depends on the closure and separation operators evaluated at the sharp-interface profile.

Proof sketch. The leading-order Γ-convergence is the standard Modica–Mortola result adapted to graphs [1]. The key ingredients are:

Compactness: Bounded energy implies BV-type regularity (the graph total variation is controlled).

Lower bound: For any sequence $u_\varepsilon \to u$ in L^1, the Cauchy–Schwarz inequality gives

$$\widetilde{\mathcal{E}}^{\varepsilon}_{\mathrm{bd}}(u_\varepsilon) \ge 2\sum_{\{x,y\}} N(x,y)|u_x - u_y|\sqrt{W(u_x)} \xrightarrow[\varepsilon\to 0]{} \sigma_{\mathrm{AC}}|\partial\{u = 1\}|_G.$$

Upper bound: A recovery sequence is constructed using the optimal profile $q(t) = \frac{1}{2}(1 + \tanh(t/2))$ adapted to the graph metric.

The self-referential corrections are lower-order because $\mathcal{E}_{\mathrm{cl}}$ and $\mathcal{E}_{\mathrm{sep}}$ are bounded on $[0, 1]^n$ and do not scale with $1/\varepsilon$. Their contribution to the surface tension can be computed by evaluating the operators at the optimal profile. □

Remark 6. *In the Γ-limit ($\varepsilon \to 0$), the surface tension is exactly σ_{AC}: the self-referential terms $\mathcal{E}_{\mathrm{cl}}$ and $\mathcal{E}_{\mathrm{sep}}$ are $O(1)$ while $\mathcal{E}_{\mathrm{bd}}$ is $O(1/\varepsilon)$, so their relative contribution vanishes. At finite ε, the effective surface tension is $\sigma_{\mathrm{eff}}(\varepsilon) = \sigma_{\mathrm{AC}} + O(\varepsilon)$. The self-referential corrections thus influence the* finite-resolution *structure of the interface (boundary width, closure*

defect, distinction profile) rather than the asymptotic surface tension.

Theorem 8 (Resolvent Co-Belonging)**.** *The resolvent* $\mathbf{C} = (I - \alpha W_{\mathrm{sym}})^{-1}$ *with* $\alpha\,\rho(W_{\mathrm{sym}}) < 1$ *satisfies:*

- **C1** Dependence: $\mathbf{C}(x, y)$ *depends on the cohesion field* u *and adjacency* N *(by construction).*
- **C2** Distinction from adjacency: $\mathbf{C}$ *achieves* $3+$ *orders of magnitude discrimination between within-formation and cross-boundary site pairs.*
- **C3″** Local monotonicity: $\mathbf{C}(x, x)$ *is monotone increasing in* $u(x)$ *with other field values held fixed.*
- **C4** Symmetry: $\mathbf{C}(x, y) = \mathbf{C}(y, x)$.

Proof sketch. **C1:** By construction, W_{sym} depends on u and N.

C2: Consider a formation-structured field on a grid graph with high-cohesion interior and low-cohesion exterior. The cohesion-weighted adjacency $W_{\mathrm{sym}}(x, y) \propto \sqrt{u(x)u(y)}N(x, y)$ effectively severs cross-boundary connections (where one of $u(x), u(y)$ is near zero). The Neumann series $\sum \alpha^k W_{\mathrm{sym}}^k$ then concentrates within the formation: paths crossing the boundary are exponentially suppressed. Numerical verification on 5×5 grids confirms within-formation values $\mathbf{C}(x, y) \sim 10^0$–$10^1$ versus cross-boundary values $\sim 10^{-3}$–10^{-4}.

C3″: Increasing $u(x)$ increases $W_{\mathrm{sym}}(x, y)$ for all neighbors y of x (via the $\sqrt{u(x)}$ factor). By the Neumann series representation, all terms $\alpha^k (W_{\mathrm{sym}}^k)_{xx}$ are non-decreasing in $u(x)$, so their sum $\mathbf{C}(x, x)$ is non-decreasing.

C4: W_{sym} is symmetric by construction, so $(W_{\mathrm{sym}}^k)_{xy} = (W_{\mathrm{sym}}^k)_{yx}$, and hence $\mathbf{C}(x, y) = \mathbf{C}(y, x)$. □

Remark 7 (C3″ gap)**.** *The C3″ proof has a subtle gap: the symmetrization step involves degree normalization* d_x *that itself depends on* $u(x)$*, and the monotonicity of* $d_x^{-1/2}$ *with respect to* $u(x)$ *requires verification. The Neumann series argument is highly plausible and is supported by numerical evidence, but the formal verification of this step is incomplete.*

We record an additional structural result connecting the separation predicate and energy.

Proposition 9 (Sep Covariance Identity)**.** *For the* $\mathbf{C}$*-weighted separation predicate, we have*

$$\mathrm{Sep}_{\mathrm{new}} = \bar{D} + \frac{n}{S}\,\mathrm{Cov}_{\mathrm{unif}}(\mathbf{C}(\cdot,\cdot), \mathbf{D}(\cdot)), \tag{27}$$

where $\bar{D} = \frac{1}{n}\sum_x \mathbf{D}(x; 1-u)$*,* $S = \sum_x \mathbf{C}(x, x)$*, and the covariance is with respect to the uniform measure over sites. In particular, for formation-structured fields (where high-cohesion sites have both high* $\mathbf{C}(x, x)$ *and high* $\mathbf{D}(x; 1-u)$*), the covariance is positive, giving* $\mathrm{Sep}_{\mathrm{new}} \geq \bar{D}$.

We now prove a quantitative lower bound on Bind at constrained energy minimizers. This is the first result connecting the variational structure (KKT conditions) directly to a diagnostic predicate.

Theorem 10 (Bind Lower Bound at Constrained Minimizers)**.** *Let* $G = (X, N)$ *be a finite connected graph with* $n = |X|$ *vertices, and let* $\hat{u}$ *be a global minimizer of* $\mathcal{E} = \lambda_{\mathrm{cl}}\mathcal{E}_{\mathrm{cl}} + \lambda_{\mathrm{sep}}\mathcal{E}_{\mathrm{sep}} + \lambda_{\mathrm{bd}}\mathcal{E}_{\mathrm{bd}}$ *on* Σ_m *satisfying strict interiority:* $0 < \hat{u}_i < 1$ *for all* i*. Let* $r = \mathrm{Cl}(\hat{u}) - \hat{u}$ *be the closure residual,* $r_T = \Pi_T(r)$ *its tangential component, and* $\bar{r}_0 = |\mathbf{1}^T r|/n$ *the per-site mean residual. Then:*

- **(a)** (Tangential residual bound.) *The tangential component satisfies*

$$\|r_T\|_2 \leq \frac{\lambda_{\mathrm{sep}} G_{\mathrm{sep}} + \lambda_{\mathrm{bd}} G_{\mathrm{bd}}}{2\lambda_{\mathrm{cl}}(1 - a_{\mathrm{cl}}/4)} + \frac{(1 + a_{\mathrm{cl}}/4)\sqrt{n}\,\bar{r}_0}{1 - a_{\mathrm{cl}}/4}, \tag{28}$$

where $G_{\mathrm{sep}} = \|\Pi_T(\nabla\mathcal{E}_{\mathrm{sep}}(\hat{u}))\|_2$ *and* $G_{\mathrm{bd}} = \|\Pi_T(\nabla\mathcal{E}_{\mathrm{bd}}(\hat{u}))\|_2$ *are projected gradient norms and* Π_T *projects onto* $T(\Sigma_m) = \{v : \sum_i v_i = 0\}$.

- **(b)** (Bind bound.)

$$\mathrm{Bind}(\hat{u}) \geq 1 - \sqrt{\frac{\|r_T\|_2^2}{n} + \bar{r}_0^2}. \tag{29}$$

- **(c)** (Substituting universal gradient bounds.) *With* $G_{\mathrm{bd}} \leq (4\alpha\lambda_{\max}(L) + \frac{2\beta}{3\sqrt{3}})\sqrt{n}$ *and* $G_{\mathrm{sep}} \leq (1 + \|J_D\|_{\mathrm{op}})\sqrt{n}$*:*

$$\mathrm{Bind}(\hat{u}) \geq 1 - \frac{\lambda_{\mathrm{bd}}(4\alpha\lambda_{\max}(L) + \frac{2\beta}{3\sqrt{3}}) + \lambda_{\mathrm{sep}}(1 + \|J_D\|_{\mathrm{op}})}{2\lambda_{\mathrm{cl}}(1 - a_{\mathrm{cl}}/4)} \tag{30}$$

up to the $\bar{r}_0$ *correction. This bound is independent of* n *when all parameters are* $O(1)$.

Proof. **Step 1: KKT and projection.** Since $\hat{u}$ minimizes $\mathcal{E}$ on Σ_m with no active box constraints, the KKT conditions give $\nabla\mathcal{E}(\hat{u}) = \nu\mathbf{1}$ for some $\nu \in \mathbb{R}$. The orthogonal projection $\Pi_T(v) = v - \frac{1}{n}(\mathbf{1}^T v)\mathbf{1}$ kills the Lagrange multiplier:

$$\lambda_{\mathrm{cl}}\,\Pi_T(\nabla\mathcal{E}_{\mathrm{cl}}) + \lambda_{\mathrm{sep}}\,\Pi_T(\nabla\mathcal{E}_{\mathrm{sep}}) + \lambda_{\mathrm{bd}}\,\Pi_T(\nabla\mathcal{E}_{\mathrm{bd}}) = 0. \tag{P-KKT}$$

Step 2: Restricted operator A_T**.** The closure gradient is $\nabla\mathcal{E}_{\mathrm{cl}} = -2(I - J_{\mathrm{Cl}})^T r$. Decompose $r = r_T + \frac{\mathbf{1}^T r}{n}\mathbf{1}$ and apply Π_T to obtain

$$\Pi_T\big((I - J_{\mathrm{Cl}})^T r\big) = A_T r_T + \frac{\mathbf{1}^T r}{n}\,\Pi_T\big((I - J_{\mathrm{Cl}})^T\mathbf{1}\big),$$

where $A_T := \Pi_T(I - J_{\mathrm{Cl}})^T|_T$ is the restricted operator on the tangent space T.

Step 3: Minimum singular value of A_T**.** The closure Jacobian satisfies $J_{\mathrm{Cl}} = \mathrm{diag}(s) \cdot M$ where $s_i = \sigma'(z_i) \cdot a_{\mathrm{cl}} \in [0, a_{\mathrm{cl}}/4]$ (since $\max\sigma' = 1/4$) and $M = (1-\eta)I + \eta P$ with $\|M\|_2 \leq 1$. For any vector w:

$$\|J_{\mathrm{Cl}} w\|_2 = \|\mathrm{diag}(s)Mw\|_2 \leq \|s\|_\infty \|Mw\|_2 \leq \tfrac{a_{\mathrm{cl}}}{4}\|w\|_2.$$

For any unit $v \in T$, using $\Pi_T v = v$ and Cauchy–Schwarz:

$$\|A_T v\|_2 \geq \langle A_T v, v\rangle = \langle (I - J_{\mathrm{Cl}})^T v, v\rangle = 1 - v^T J_{\mathrm{Cl}} v \geq 1 - \|J_{\mathrm{Cl}}\|_2 \geq 1 - a$$

Hence $\sigma_{\min}(A_T) \geq 1 - a_{\mathrm{cl}}/4 > 0$ and A_T is invertible on T.

Step 4: Assembly. Substituting into (P-KKT), the triangle inequality gives

$$\|A_T r_T\|_2 \leq \frac{\lambda_{\mathrm{sep}} G_{\mathrm{sep}} + \lambda_{\mathrm{bd}} G_{\mathrm{bd}}}{2\lambda_{\mathrm{cl}}} + \frac{|\mathbf{1}^T r|}{n}\|\Pi_T((I - J_{\mathrm{Cl}})^T\mathbf{1})\|_2.$$

The second term is bounded by $(1 + a_{\mathrm{cl}}/4)\sqrt{n}\,\bar{r}_0$ since $\|(I - J_{\mathrm{Cl}})^T\mathbf{1}\|_2 \leq (1 + a_{\mathrm{cl}}/4)\sqrt{n}$. Dividing by $\sigma_{\min}(A_T) \geq 1 - a_{\mathrm{cl}}/4$ yields (28). The Bind bound (29) follows from $\|r\|_2^2 = \|r_T\|_2^2 + (\mathbf{1}^T r)^2/n$ and the definition $\mathrm{Bind} = 1 - \|r\|_2/\sqrt{n}$.

Step 5: Gradient bounds. For $\mathcal{E}_{\rm bd}$: $\nabla\mathcal{E}_{\rm bd} = 4\alpha L\hat{u} + \beta W'(\hat{u})$ with $\|4\alpha L\hat{u}\|_2 \le 4\alpha\lambda_{\max}(L)\sqrt{n}$ (since $\hat{u}_i \in [0,1]$) and $\|\beta W'(\hat{u})\|_2 \le \frac{2\beta}{3\sqrt{3}}\sqrt{n}$ (the maximum of $|W'(u)| = |2u(1-u)(1-2u)|$ on $[0,1]$ is $\frac{2}{3\sqrt{3}}$). For $\mathcal{E}_{\rm sep}$: $\nabla\mathcal{E}_{\rm sep} = (1-D) - J_D^T\hat{u}$ with $\|J_D\|_2 \le a_D(1+\lambda_D)/4$. Since Π_T does not increase norms, the substitution gives (30). □

Corollary 11 (Asymptotic Bind Satisfaction)**.** *For any $\varepsilon > 0$, there exists $\Lambda(\varepsilon, graph, params) > 0$ such that if $\lambda_{\rm cl} > \Lambda \cdot (\lambda_{\rm bd} + \lambda_{\rm sep})$, then $\mathrm{Bind}(\hat{u}) \ge 1 - \varepsilon$, provided $\bar{r}_0 < \varepsilon$.*

Remark 8 (Structural role of the mean residual)**.** *The bound has two components with fundamentally different character. The tangential part $\|r_T\|_2^2/n$ is fully controlled by the KKT structure: it is $O((\lambda_{\rm bd} + \lambda_{\rm sep})^2/\lambda_{\rm cl}^2)$ and vanishes under closure dominance. The mean residual $\bar{r}_0 = |\sum_i(\mathrm{Cl}(\hat{u})_i - \hat{u}_i)|/n$ is a structural property of the closure operator—it measures the per-site mass mismatch between $\mathrm{Cl}(\hat{u})$ and $\hat{u}$, which the projected KKT cannot control without circularity (the Lagrange multiplier ν absorbs the mean-direction information). Empirically, $\bar{r}_0 < 0.02$ at all tested formations, because well-formed formations are approximately binary: interior sites have $\mathrm{Cl}(\hat{u})_i \approx \hat{u}_i \approx 1$ and exterior sites have $\mathrm{Cl}(\hat{u})_i \approx \hat{u}_i \approx 0$, so the mass mismatch is concentrated at the narrow interface. A rigorous bound on $\bar{r}_0$ at energy minimizers requires a quantitative binary-approximation result at finite parameters, which remains an open problem.*

Remark 9 (Explicit constants for grid graphs)**.** *For a 2D grid with $d_{\max} = 4$, $\lambda_{\max}(L) \le 2d_{\max} = 8$, and default parameters ($a_{\rm cl} = 3.5$, $\alpha = 1$, $\beta = 10$, $a_D = 5$, $\lambda_D = 1$), the bound* (30) *gives* $\mathrm{Bind} \ge 1 - (35.85\lambda_{\rm bd} + 3.5\lambda_{\rm sep})/(0.25\lambda_{\rm cl})$. *This is non-vacuous when $\lambda_{\rm cl} \gg 143\lambda_{\rm bd} + 14\lambda_{\rm sep}$, which quantifies the closure dominance condition.*

IV. Self-Referential Optimal Transport

The static theory of Section III characterizes formations at a single time. We now develop the temporal dimension: how formations persist through time via self-referential transport.

A. The Cohesion Fingerprint

The key construction is a *cohesion fingerprint* that encapsulates the operator pair at each site:

$$\phi(x) = \big(u(x),\ \mathrm{Cl}(u)(x),\ \mathbf{D}(x; 1-u)\big) \in [0,1]^3. \tag{31}$$

The three components are: raw cohesion, self-completion, and self-contrast. The resolvent diagonal $\mathbf{C}(x,x)$ is demoted from the canonical fingerprint: it contributes $< 0.4\%$ of the fingerprint gap but has Jacobian norm $\sim$9300, making the cost matrix poorly conditioned. Every component depends only on $(u, \mathbf{N})$—no external features are required.

B. Self-Referential Transport Cost

Given cohesion fields u_t, u_s at times t, s, we define the transport cost:

$$c(x,y) = \frac{d_G(x,y)^2}{2\sigma^2} + \gamma\|\phi_t(x) - \phi_s(y)\|^2, \tag{32}$$

where d_G is the graph distance, $\sigma > 0$ controls spatial tolerance, $\gamma > 0$ weights fingerprint similarity, and ϕ_t, ϕ_s are the fingerprints at times t and s respectively.

This cost is *self-referential*: it depends on the fields u_t, u_s through the fingerprints. The optimal transport plan depends on the cost, which depends on the fields, which are updated using the transport plan. This creates a fixed-point problem in transport-plan space.

C. The Fixed-Point Formulation

The self-referential transport problem is formulated as follows. Given u_t at time t, define the map $\Phi : \Sigma_m \to \Sigma_m$ by:

1) Compute fingerprints $\phi_t(x)$ from u_t and $\phi_s(y)$ from a candidate u_s.
2) Compute the cost matrix $c(x,y)$ from (32).
3) Solve the entropy-regularized optimal transport problem:
$$M^* = \arg\min_{\substack{M\ge 0\\ M\mathbf{1}\le\mathbf{1}}} \sum_{x,y} M(x,y)\,c(x,y) + \varepsilon_{\rm OT}\sum_{x,y} M(x,y)\log M(x,y) \tag{33}$$
4) Compute the transported field and minimize the static energy at time s initialized from the transported data, yielding the updated u_s.

A fixed point of Φ is a field u_s^* such that $\Phi(u_s^*) = u_s^*$: the transport plan, the fingerprints, and the field are mutually consistent.

Theorem 12 (Transport Fixed-Point Existence)**.** *For any $\lambda_{\rm tr} > 0$ and $\varepsilon_{\rm OT} > 0$, the self-referential transport map $\Phi : \Sigma_m \to \Sigma_m$ has at least one fixed point.*

Proof. The proof uses Schauder's fixed-point theorem via finite-time flow truncation.

Step 1 (Truncated map). Define $\Phi_\tau : \Sigma_m \to \Sigma_m$ by running projected gradient descent for finite time τ from the transported field. Φ_τ is continuous: entropic OT gives C^∞ transport plans; finite-time gradient flow is continuous in initial conditions.

Step 2 (Fixed points). By Schauder's theorem (Φ_τ continuous on compact convex Σ_m), each Φ_τ has a fixed point $u_s^{(\tau)}$.

Step 3 (Passage to $\tau \to \infty$). By compactness, $\{u_s^{(\tau_k)}\}$ has a convergent subsequence with limit u_s^*. The Łojasiewicz convergence theorem (Theorem 5) ensures $\Phi(u_s^*) = u_s^*$. □

Remark 10 (Computational validation)**.** *In the weak transport regime ($\lambda_{\rm tr} = 0.1$, $\varepsilon_{\rm OT} = 1.0$), alternating minimization converges in 1–3 iterations on grids up to 50×50, consistent with the contraction analysis. The cohesion fingerprint $\varphi(x) = (u(x), \mathrm{Cl}(u)(x), D(x; 1-u))$ produces cost matrices that yield core-to-core transport concentration exceeding 95% at deep interior sites. The transport-based persistence predicate* $\mathsf{Persist}_{\rm tr} \approx 0.99$ *for statically equivalent formations, confirming temporal identity tracking in the weak regime.*

Remark 11 (Uniqueness and multiple fixed points)**.** *Uniqueness of the fixed point is* not *guaranteed in general. Multiple fixed points correspond to different identity assignments—different ways of matching the formation at time t to a formation at time s. We identify three regimes:*

1) ***Weak transport*** *($\lambda_{\rm tr}$ small, $\varepsilon_{\rm OT}$ large): The map Φ is a contraction, giving a unique fixed point. The contraction follows because large $\varepsilon_{\rm OT}$ smooths the OT solution (reducing sensitivity to cost perturbations) while small $\lambda_{\rm tr}$ limits the feedback from the transported field to the fingerprints. Alternating minimization converges.*
2) ***Moderate and strong transport****: Comprehensive experiments sweeping $\lambda_{\rm tr} \in [0.01, 10]$ with five different initializations on grids up to 15×15 find* no multiplicity*: all initializations converge to the same field. The re-optimization step acts as a discrete attractor, collapsing the continuous transport output to a finite set of energy minima.*
3) ***Analytical uniqueness****: The transport confinement bound provides a sufficient condition for uniqueness: $C_{\rm conf}\sqrt{m} < r_{\rm basin}$, replacing the contraction condition* (34) *with a weaker, formation-geometry-dependent criterion.*

Remark 12 (Novelty in optimal transport)**.** *The self-referential transport formulation—where the cost depends on the fields being transported—is, to our knowledge, without direct precedent in the optimal transport literature. The closest relative is mean-field game theory [7], where the cost depends on the* distribution *of agents. But even in mean-field games, the cost does not depend on the* specific values *being transported. Our formulation creates a fixed-point equation in transport-plan space that is structurally distinct from any existing OT problem.*

D. Temporal Persistence Results

Using the self-referential transport framework, we state a temporal persistence result. Three of six gaps identified by audit have been closed; three remain open (see proof sketch below).

Definition 1 (ε-gentle transition)**.** *Let $X = X_t = X_s$ be a fixed support space. A transition from t to s is* ***ε-gentle*** *if:* (G1) $\|\mathcal{E}_s - \mathcal{E}_t\|_{C^2(\Sigma_m)} \le \varepsilon_1$ *(energy landscape closeness);* (G2) $\|\mathbf{M}_{t\to s}\hat{u}_t - \hat{u}_t\|_2 \le \varepsilon_2$ *(transport displacement);* (G3) $\|\mathbf{N}_s - \mathbf{N}_t\|_{\rm op} \le \varepsilon_3$ *(adjacency closeness); where* $\varepsilon = \max(\varepsilon_1, \varepsilon_2, \varepsilon_3)$.

Theorem 13 (Temporal Core Inheritance)**.** *Let $X = X_t = X_s$ with $n = |X|$. Let $\hat{u}_t \in \Sigma_m$ be a formation-structured local minimizer of $\mathcal{E}_t$ on Σ_m with constrained Hessian spectral gap $\mu > 0$. Suppose the transition from t to s is ε-gentle (Definition 1) with $\varepsilon_1 < \mu/2$. Then:*

(a) (Minimizer persistence.) *$\mathcal{E}_s$ has a local minimizer $\hat{u}_s \in \Sigma_m$ with $\|\hat{u}_s - \hat{u}_t\|_2 \le 2\varepsilon_1/\mu$, non-degenerate with spectral gap $\ge \mu/2$.*
(b) (Gradient flow convergence—conditional on basin containment.) *If additionally $2\varepsilon_2 + 2\varepsilon_1/\mu < r_{\rm basin}$ where $r_{\rm basin} \ge 0.210$ (from the β-independent energy barrier estimate; see Remark 15), then the projected gradient flow from $\pi_\Sigma(\mathbf{M}_{t\to s}\hat{u}_t)$ converges exponentially to $\hat{u}_s$ by Theorem 5.*
(c) (Core inclusion.) $\mathrm{Core}_t(\hat{u}_t) \subseteq \mathrm{Core}_s(\hat{u}_s, \theta_{\rm core} - 2\varepsilon_1/\mu)$.
(d) (Exact threshold preservation.) *If the interior gap satisfies $\min_{x\in\mathrm{Core}_t}(\hat{u}_t(x) - \theta_{\rm core}) > 2\varepsilon_1/\mu$ (equivalently, $\varepsilon_1 < \mu\gamma_{\rm int}/2$), then $\mathrm{Core}_t(\hat{u}_t) \subseteq \mathrm{Core}_s(\hat{u}_s, \theta_{\rm core})$.*

Remark 13 (Exact threshold preservation via interior gap)**.** *The double-well structure of $\mathcal{E}_{\rm bd}$ forces exponential saturation of interior field values (Proposition 14), giving an interior gap $\ge (1-\theta_{\rm core}) - O(\exp(-c_0\,\delta)) - O(1/\beta)$, where $c_0 = \mathrm{arccosh}(1+\kappa^2/d_{\min})$ grows logarithmically in β/α, for core sites at graph distance $\delta \ge 2$ from the formation boundary. This makes part (d) satisfiable for ε_1 sufficiently small relative to μ.*

Proposition 14 (Interior Gap Lower Bound)**.** *Let $\hat{u} \in \Sigma_m$ be a constrained minimizer of $\mathcal{E} = \lambda_{\rm cl}\mathcal{E}_{\rm cl} + \lambda_{\rm sep}\mathcal{E}_{\rm sep} + \lambda_{\rm bd}\mathcal{E}_{\rm bd}$ with $\beta/\alpha > 4\lambda_2/|W''(c)|$ (phase-separated regime). Assume:*

(H2) Core depth: $\delta_{\min} := \min_{x\in\mathrm{Core}} d_G(x, \partial\,\mathrm{Core}) \ge 2$.
(H3) Phase separation: $\beta > 11\alpha$ *(for default $\theta_{\rm core} = 0.9$, $d_{\min} = 4$).*
- *The graph has minimum degree $d_{\min} \ge 2$.*

Define the screening parameter $\kappa = \sqrt{\beta/(2\alpha)}$, the per-hop screening decay rate $c_0 = \mathrm{arccosh}(1+\kappa^2/d_{\min})$, the operator constant $C_{\rm op} \le 4(1 + a_{\rm cl} + a_D)$, and the weight ratio $R = \max(\lambda_{\rm cl}/\lambda_{\rm bd},\, \lambda_{\rm sep}/\lambda_{\rm bd})$.

Then for any $x \in \mathrm{Core}(\hat{u})$ with depth $\delta(x) := d_G(x, \partial\,\mathrm{Core})$:

$$\hat{u}(x) \ge \theta_{\rm core} + (1-\theta_{\rm core}) - 2\exp(-c_0\,\delta(x)) - \frac{C_{\rm op}\,R}{\beta}. \quad (34)$$

Two-tier consequence:

- Deep core *($\delta \ge 2$): gap $\ge (1-\theta_{\rm core}) - O(\exp(-2c_0)) - O(1/\beta) \to 1-\theta_{\rm core}$.*
- Shallow core *($\delta = 1$): gap $\approx O(1/\sqrt{\beta})$.*

Proof. The proof proceeds in four steps: linearization at the well $u \approx 1$, exponential decay via the discrete screened Laplacian, KKT perturbation bounds for operator corrections, and assembly of explicit constants.

Step 1: Linearization. Set $v = 1 - \hat{u}$ on the core interior. At the constrained minimizer, the KKT conditions give $\nabla\mathcal{E}(\hat{u}) = \nu\,\mathbf{1}$ on unconstrained (interior box) coordinates, where ν is the volume Lagrange multiplier. The boundary energy's Euler–Lagrange equation at site x reads

$$\lambda_{\rm bd}\big[4\alpha(L\hat{u})_x + \beta W'(\hat{u}_x)\big] + \lambda_{\rm cl}\,\partial_{u_x}\mathcal{E}_{\rm cl} + \lambda_{\rm sep}\,\partial_{u_x}\mathcal{E}_{\rm sep} = \nu.$$

For core sites where $\hat{u}_x \approx 1$, we have $v_x \ll 1$. The double-well derivative linearizes as $W'(u) = 2u(1-u)(1-2u) \approx -2v_x + O(v_x^2)$, and the graph Laplacian satisfies $(L\hat{u})_x = d_x\hat{u}_x - \sum_{y\sim x}\hat{u}_y = -(Lv)_x$. Substituting and retaining leading-order terms in v:

$$\lambda_{\rm bd}\big[-4\alpha(Lv)_x - 2\beta\,v_x\big] + f_x = \nu,$$

where $f_x := \lambda_{\rm cl}\,\partial_{u_x}\mathcal{E}_{\rm cl} + \lambda_{\rm sep}\,\partial_{u_x}\mathcal{E}_{\rm sep}$ collects operator corrections. Rearranging yields the discrete screened Poisson equation

$$\sum_{y\sim x} v_y \approx \Big(d_x + \kappa^2\Big)v_x + g_x, \qquad \kappa^2 := \frac{\beta}{2\alpha}, \quad (35)$$

where $g_x = (\nu - f_x)/(4\alpha\lambda_{\rm bd})$ absorbs the Lagrange multiplier and operator corrections. For deep interior sites where $v \approx 0$, the multiplier ν is $O(v)$ (from the volume-dominated balance), so g_x is controlled by f_x.

Step 2: Exponential decay via lattice Green's function. The discrete screened Laplacian $-\Delta_G + \kappa^2 I$ (where $(\Delta_G v)_x = \sum_{y\sim x}(v_y - v_x)$ is the graph Laplacian acting on v) has a Green's function $G_\kappa(x,y)$ that decays exponentially:

$$G_\kappa(x,y) \le C\exp\bigl(-c_0\, d_G(x,y)\bigr),$$

where $c_0 = \operatorname{arccosh}(1+\kappa^2/d_{\min})$ is the per-hop decay rate of the discrete screened Laplacian. For large $\kappa^2/d_{\min}$, $c_0 \approx \ln(2\kappa^2/d_{\min})$, growing logarithmically in β/α. (On a 2D square grid with $d_{\min} = 4$ and $\kappa^2 = 25$ ($\beta = 50\alpha$): $c_0 = \operatorname{arccosh}(7.25) \approx 2.67$.)

By the discrete maximum principle applied to (36), the deviation v_x at a core site x is controlled by the boundary values $v|_{\partial\,\mathrm{Core}}$ and the source g:

$$v(x) \le \max_{y\in\partial\,\mathrm{Core}} v(y)\cdot\exp\bigl(-c_0\,\delta(x)\bigr) + \text{source contribution}.$$

Since $v(y) = 1 - \hat u(y) \le 1$ at $\partial\,\mathrm{Core}$, the boundary contribution is at most $\exp(-c_0\,\delta(x))$. The factor of 2 in (35) accounts for the maximum principle constant when converting from Green's function bounds to pointwise estimates.

Step 3: KKT perturbation bound for operator corrections. The source term g_x in (36) is dominated by the operator gradients f_x. These are bounded component-wise:

- $|\partial_{u_x}\mathcal{E}_{\rm cl}| \le 2(1 + a_{\rm cl}/4)$, since the closure Jacobian satisfies $\|J_{\rm Cl}\|_{\rm op} \le a_{\rm cl}/4$ by axiom A3.
- $|\partial_{u_x}\mathcal{E}_{\rm sep}| \le 2$, since distinction values lie in $[0,1]$.

Combining: $|f_x| \le \max(\lambda_{\rm cl},\lambda_{\rm sep})\cdot 2(2 + a_{\rm cl}/4 + a_D) \le C_{\rm op}\cdot\max(\lambda_{\rm cl},\lambda_{\rm sep})$ with $C_{\rm op} \le 4(1 + a_{\rm cl} + a_D)$. In the screened Poisson equation, this source is suppressed by $\kappa^2 = \beta/(2\alpha)$: the correction to v_x is at most $|f_x|/(\lambda_{\rm bd}\kappa^2) = C_{\rm op}R/(2\alpha\kappa^2) = C_{\rm op}R/\beta$. Numerically, the closure and separation gradients contribute $< 5\%$ of the double-well force at core sites (verified computationally at default parameters).

Step 4: Assembly. Combining Steps 2–3, the deviation satisfies

$$v(x) = 1 - \hat u(x) \le 2\exp\bigl(-c_0\,\delta(x)\bigr) + \frac{C_{\rm op}\,R}{\beta},$$

hence $\hat u(x) \ge 1 - 2\exp(-c_0\,\delta(x)) - C_{\rm op}\,R/\beta$. Subtracting $\theta_{\rm core}$ yields (35). For deep core ($\delta \ge 2$), the exponential term is $O(\exp(-2c_0))$, which falls below 10^{-3} at $\beta \ge 50$ on a 2D grid ($c_0 = \operatorname{arccosh}(7.25) \approx 2.67$). For boundary core ($\delta = 1$), the transition layer analysis gives the coarser $O(1/\sqrt{\beta})$ bound from the discrete profile matching at the formation edge.

Hypotheses discussion. The literal condition $\delta_{\min} \ge 2$ (every core site at distance ≥ 2 from non-core) fails on finite grids because boundary core sites always neighbor non-core sites. The operationally correct reading of (H2) is *deep core non-emptiness*: $\{x \in \mathrm{Core} : \delta(x) \ge 2\} \ne \emptyset$, since the concentration bound (35) is applied per-site for $\delta(x) \ge 2$. A systematic sweep (Experiment 13: 4 grid sizes 8^2–20^2, 5 values of β, 4 volume fractions, 4 closure strengths; 240 combinations) confirms deep core non-emptiness in 208/219 formations with non-empty core (95.0%). All 11 failures occur at weak phase separation ($\beta \le 10$, $a_{\rm cl} \le 3.0$) with small cores (≤ 27 sites); at $\beta \ge 20$ the deep core exists universally (144/144) and contains a median 70.6% of core mass. Condition (H3) ($\beta > 11\alpha$ for $d_{\min} = 4$) ensures the exponential and $O(1/\beta)$ terms are jointly smaller than $1 - \theta_{\rm core} = 0.1$. □

Proposition 15 (Basin Radius via Energy Sublevel Set)**.** *Let $\hat u \in \Sigma_m$ be a local minimizer of $\mathcal{E}$ on Σ_m with:*

(B1) Non-degeneracy. *Active-constraint-aware spectral gap $\mu_{\mathcal{F}} > 0$, where $\mathcal{F} = \{i : 0 < \hat u_i < 1\}$ is the set of free variables.*

(B2) Energy barrier. *There exists $\Delta > 0$ such that every critical point u^* of $\mathcal{E}|_{\Sigma_m}$ with $\mathcal{E}(u^*) \le \mathcal{E}(\hat u) + \Delta$ and $u^* \ne \hat u$ lies outside the connected component of $\{\mathcal{E} < \mathcal{E}(\hat u) + \Delta\}$ containing $\hat u$.*

Then the basin of attraction of $\hat u$ under the projected gradient flow on Σ_m contains $B(\hat u, r_{\rm basin}) \cap \Sigma_m$ where

$$r_{\rm basin} = \sqrt{\frac{2\Delta}{\lambda_{\max}}}, \tag{36}$$

with $\lambda_{\max}$ the largest eigenvalue of $\nabla^2_\Sigma\mathcal{E}(\hat u)$ on $T_{\hat u}\Sigma_m$.

Proof. Step 1 (Forward invariance). The projected gradient flow $\dot u = -\Pi_\Sigma\nabla\mathcal{E}(u)$ (with projection onto the tangent cone accounting for box constraints) satisfies $\frac{d}{dt}\mathcal{E}(u(t)) = -\|\Pi_\Sigma\nabla\mathcal{E}(u(t))\|^2 \le 0$, so the sublevel set $S_\Delta := \{u \in \Sigma_m : \mathcal{E}(u) < \mathcal{E}(\hat u) + \Delta\}$ is forward-invariant.

Step 2 (Unique critical point in component). Let $\mathcal{C}$ be the connected component of S_Δ containing $\hat u$. By (B2), $\mathcal{C}$ contains no other critical point of $\mathcal{E}|_{\Sigma_m}$.

Step 3 (Convergence by Łojasiewicz). Any trajectory $u(t)$ starting in $\mathcal{C}$ remains in $\mathcal{C}$ (forward invariance) with monotone decreasing energy bounded below by $\mathcal{E}(\hat u)$. The ω-limit set is non-empty (compactness of $\overline{\mathcal{C}} \subset \Sigma_m \cap [0,1]^n$), connected, and consists of critical points (LaSalle invariance principle). Since $\hat u$ is the only critical point in $\mathcal{C}$, $\omega(u) = \{\hat u\}$. By the Łojasiewicz convergence theorem (Theorem 5—the energy is real-analytic since $b_D = 0$), convergence is exponential at rate $\mu_{\mathcal{F}}/2$ (Łojasiewicz exponent $\theta = 1/2$ at non-degenerate minimizers).

Step 4 (Ball containment). For $\|u_0 - \hat u\|_2 \le r_{\rm basin}$, Taylor expansion gives $\mathcal{E}(u_0) \le \mathcal{E}(\hat u) + \frac{\lambda_{\max}}{2}r_{\rm basin}^2 = \mathcal{E}(\hat u) + \Delta$, so $u_0 \in S_\Delta$. The ball $B(\hat u, r_{\rm basin})$ is connected and intersects $\mathcal{C}$ at $\hat u$, so $u_0 \in \mathcal{C}$. By Step 3, the flow from u_0 converges to $\hat u$. □

Remark 14 (β-independent basin radius)**.** *For the double-well energy in the phase-separated regime, the energy barrier per core node scales as $\Delta_{\rm node} \ge \beta[W(c) - W(0)] \ge 0.0441\beta$ (at $c = 0.3$, since $W(0.3) = 0.3^2 \times 0.7^2 = 0.0441$), while $\lambda_{\max} \le 2\beta +$ lower-order terms (from $W''(0) = W''(1) = 2$). The β factors cancel:*

$$r_{\rm basin} \ge \sqrt{\frac{2\cdot 0.0441\beta}{2\beta}} = \sqrt{0.0441} \approx 0.210.$$

Subsequent analysis reveals this bound is too optimistic: the isotropic basin radius is controlled by the boundary-mode barrier $\Delta_{\rm bdy}$, not $\Delta_{\rm node}$. The ellipsoidal basin provides 1.5–3.3× larger radii transverse to the soft mode. Near bifurcation ($\mu \to 0$), $\Delta_{\rm bdy} = O(\mu^3)$, but the directional extension recovers full persistence for a 2.5–4.3× wider spectral gap range.

Proposition 16 (Barrier Stability Under Gentle Transition)**.** *Let $\hat{u}_t$ be a local minimizer of $\mathcal{E}_t$ with energy barrier $\Delta_t > 0$ (in the sense of Proposition 15, B2), with all barrier-defining saddle points non-degenerate (Morse condition—generic by Sard's theorem). Under an ε-gentle transition (Definition 1) with $\varepsilon_1 < \mu/2$, the IFT minimizer $\hat{u}_s$ has energy barrier*

$$\Delta_s \geq \Delta_t - 2\varepsilon_1. \tag{37}$$

Proof sketch. From (G1), $|\mathcal{E}_s(u) - \mathcal{E}_t(u)| \leq \varepsilon_1$ for all $u \in \Sigma_m$. The minimizer energy increases by at most ε_1: $\mathcal{E}_s(\hat{u}_s) \leq \mathcal{E}_t(\hat{u}_t) + \varepsilon_1$. Any index-1 saddle u_t^* of $\mathcal{E}_t$ near the barrier persists under IFT (non-degeneracy extends to saddle points), giving $\mathcal{E}_s(u_s^*) \geq \mathcal{E}_t(u_t^*) - \varepsilon_1$. Combining: $\Delta_s \geq (\mathcal{E}_t(u_t^*) - \varepsilon_1) - (\mathcal{E}_t(\hat{u}_t) + \varepsilon_1) = \Delta_t - 2\varepsilon_1$. □

Proposition 17 (Basin Containment)**.** *Under the hypotheses of Theorem 13, plus energy barrier $\Delta_t > 0$ with non-degenerate saddle points, if the* gentle-transition *conditions*

$$\varepsilon_1 < \frac{\Delta_t}{4}, \qquad 2\varepsilon_2 + \frac{2\varepsilon_1}{\mu} < \sqrt{\frac{\Delta_t - 2\varepsilon_1}{\lambda_{\max,s}/2}} \tag{38}$$

hold, then the projected gradient flow on Σ_m from $\tilde{u}_\Sigma := \pi_\Sigma(\mathbf{M}_{t\to s}\hat{u}_t)$ converges exponentially to $\hat{u}_s$.

Proof sketch. By Proposition 16, $\Delta_s \geq \Delta_t - 2\varepsilon_1 > \Delta_t/2 > 0$. By Proposition 15, $r_s \geq \sqrt{2\Delta_s/\lambda_{\max,s}}$. The transported-and-projected field satisfies $\|\tilde{u}_\Sigma - \hat{u}_s\|_2 \leq 2\varepsilon_2 + 2\varepsilon_1/\mu$ (IFT bound plus (G2)). Condition (39) ensures $\|\tilde{u}_\Sigma - \hat{u}_s\| < r_s$, so Proposition 15 gives convergence. □

Lemma 18 (Fingerprint Amplification)**.** *Let $\hat{u}$ be a formation-structured minimizer on $(X, \mathbf{N})$ with interior gap $\delta_{\rm int} := \min_{x\in{\rm Core}}(\hat{u}(x) - \theta_{\rm core}) > 0$ and exterior bound $\delta_{\rm ext} := \max_{x\notin{\rm Core}} \hat{u}(x) < \theta_{\rm core}$. Define the* ***fingerprint gap***

$$\Delta_\phi^2 := \min_{x\in{\rm Core}_t} \min_{z\notin{\rm Core}_s} \|\phi_t(x) - \phi_s(z)\|^2$$

where ϕ is the cohesion fingerprint (31)*. Then*

$$\Delta_\phi^2 \geq (\theta_{\rm core}+\delta_{\rm int}-\delta_{\rm ext})^2+({\rm Cl}_{\rm core}-{\rm Cl}_{\rm ext})^2+(D_{\rm core}-D_{\rm ext})^2 \tag{39}$$

where subscripts denote the operator values at core and exterior sites respectively. The operator pair $({\rm Cl}, D)$ amplifies the raw u-gap by a factor of approximately 3×.

Proof. Each component of $\phi(x)$ is a continuous function of u and the neighborhood structure. We bound each squared gap from below.

(i) u-component. For $x \in {\rm Core}_t$ and $z \notin {\rm Core}_s$: $|u_t(x) - u_s(z)| \geq (\theta_{\rm core} + \delta_{\rm int}) - \delta_{\rm ext}$ by definition.

(ii) Closure component. For the sigmoid realization ${\rm Cl}(u)(x) = \sigma(a_{\rm cl}((1-\eta)u(x) + \eta(Pu)(x) - \tau_{\rm cl}))$, the pre-activation at a core site with core neighbors satisfies $(1-\eta)u_{\rm core} + \eta\bar{u}_{\rm core} - \tau_{\rm cl} > 0$, while at an exterior site it is negative (for $\tau_{\rm cl} = 0.5$). By the mean value theorem, ${\rm Cl}_{\rm core} - {\rm Cl}_{\rm ext} \geq \sigma'_{\min} \cdot a_{\rm cl} \cdot (u_{\rm core} - u_{\rm ext})$.

(iii) Distinction component. $D(x; 1-u) = \sigma(a_D(Pu(x) - \lambda_D P(1-u)(x)) - \tau_D)$. At core sites the argument is large and positive; at exterior sites, large and negative. The gap amplifies the u-gap by sigmoid steepness a_D.

(iv) Co-belonging diagonal. $C(x,x) = [(I - \alpha_C W_{\rm sym})^{-1}]_{xx}$. Core sites have high $W_{\rm sym}$ entries giving $C(x,x) > 1$; exterior sites have $C(x,x) \approx 1$. The normalized gap is $O(\alpha_C \cdot u_{\rm core})$.

At default parameters ($\theta_{\rm core} = 0.9$, $a_{\rm cl} = 3.5$, $a_D = 5.0$): u-gap$^2 \approx 0.72$, Cl-gap$^2 \approx 0.40$, D-gap$^2 \approx 0.94$, totaling $\Delta_\phi^2 \approx 2.06$ versus the raw u-gap of 0.72—amplification of ∼3×. □

Proposition 19 (Two-Tier Transport Concentration)**.** *Let $\hat{u}_t, \hat{u}_s$ be formation-structured minimizers on a fixed support space X with $n = |X|$, and let M^* be the entropic partial OT plan with regularization $\varepsilon_{\rm OT} > 0$ and fingerprint weight $\gamma > 0$. For depth parameter $\delta > 0$, define the δ-deep core ${\rm Core}_t^\delta := \{x \in X : \hat{u}_t(x) \geq \theta_{\rm core} + \delta\}$ and the depth-dependent fingerprint gap $\Delta_\phi^2(\delta)$ via Lemma 18. Suppose:*

(TC1) The fingerprint gap satisfies $\Delta_\phi^2(\delta) > 0$ (ensured by Proposition 14 for depth ≥ 2).
(TC2) The spatial scale satisfies $\sigma^2 \geq {\rm diam}(X)^2/2$.
(TC3) The concentration condition holds: $(\gamma\Delta_\phi^2(\delta) - {\rm diam}(X)^2/(2\sigma^2))/\varepsilon_{\rm OT} > 2\log n + \log(1/\theta_{\rm core})$.

Then:

(a) (Deep core concentration.) *For every $x \in {\rm Core}_t^\delta$ with $\delta \geq 2$:*

$$\frac{\sum_{y\in{\rm Core}_s} M^*(x,y)}{\sum_{y\in X} M^*(x,y)} \geq 1 - \frac{n}{\theta_{\rm core}} \cdot \exp\left(-\frac{\gamma\Delta_\phi^2(\delta) - {\rm diam}(X)^2/(2\sigma^2}{\varepsilon_{\rm OT}}\right. \tag{40}$$

Under (TC3)*, the right-hand side exceeds $1 - 1/n$.*

(b) (Shallow core fallback.) *For boundary core sites ($\delta(x) = 1$): concentration is weak. These sites satisfy $\hat{u}_s(x) \geq \theta_{\rm core} - 2\varepsilon_1/\mu$ from Theorem 13 (c) only.*

(c) (Boundary thinness.) $|\,{\rm Core}_t \setminus {\rm Core}_t^\delta\,| \leq |\partial\,{\rm Core}_t\,|$ *for β/α sufficiently large: the shallow core is at most one graph-hop layer wide.*

Proof. The proof proceeds in five steps. Throughout, fix $x \in {\rm Core}_t^\delta$ with $\delta \geq 2$.

Step 1: Fingerprint gap lower bound. By Lemma 18 and Proposition 14, deep core sites ($\delta \geq 2$) have interior gap $\delta_{\rm int} \geq (1 - \theta_{\rm core}) - 2\exp(-c_0\delta) - C_{\rm op}R/\beta > 0$ under hypotheses (H2)–(H3). The fingerprint gap at these sites satisfies (40) with each operator component amplifying the raw u-gap. Condition (TC1) is therefore satisfied.

Step 2: Cost comparison. Fix $x \in {\rm Core}_t^\delta$, let $y^* \in {\rm Core}_s$ be a core target, and let $z \notin {\rm Core}_s$ be a non-core target. Using the transport cost (32):

$$c(x,z) - c(x,y^*) = \frac{d_G(x,z)^2 - d_G(x,y^*)^2}{2\sigma^2} + \gamma\big(\|\phi_t(x) - \phi_s(z)\|^2 - \|\phi_t(x)$$

The fingerprint term is bounded below by $\Delta_\phi^2(\delta)$: we have $\|\phi_t(x) - \phi_s(z)\|^2 \geq \Delta_\phi^2(\delta)$ by definition, and $\|\phi_t(x) - \phi_s(y^*)\|^2 \geq 0$. (The cross-time residual $\|\phi_t(x) - \phi_s(y^*)\|^2 = O(\varepsilon_1^2/\mu^2)$ from the IFT displacement bound in Theorem 13 (a) is negligible for ε_1 small.) The spatial term satisfies $(d_G(x,z)^2 - d_G(x,y^*)^2)/(2\sigma^2) \geq -\mathrm{diam}(X)^2/(2\sigma^2)$ in the worst case. Combining:

$$c(x,z) - c(x,y^*) \geq \gamma\Delta_\phi^2(\delta) - \frac{\mathrm{diam}(X)^2}{2\sigma^2} =: \Delta_c. \quad (41)$$

Under (TC2), the spatial penalty $\mathrm{diam}(X)^2/(2\sigma^2) \leq 1$, so $\Delta_c > 0$ follows from (TC3).

Step 3: Dual potential bound. The entropic partial OT with sub-stochastic constraints $\sum_y M^*(x,y) \leq u_t(x)$, $\sum_x M^*(x,y) \leq u_s(y)$ admits dual potentials f^*, g^* with $f^*(x) \leq 0$ and $g^*(y) \leq 0$ (the non-positive constraint is complementary to the $\leq$ marginal inequalities; see [3, §4.2]). The optimal plan takes the form

$$M^*(x,y) = \exp\left(\frac{f^*(x) + g^*(y) - c(x,y)}{\varepsilon_{\mathrm{OT}}}\right).$$

Since $f^*(x) + g^*(y) \leq 0$, we obtain the entry-wise bound $M^*(x,y) \leq \exp(-c(x,y)/\varepsilon_{\mathrm{OT}})$. For $x \in \mathrm{Core}_t^\delta$ and $z \notin \mathrm{Core}_s$, the cost gap (42) gives $c(x,z) \geq \Delta_c$, so:

$$M^*(x,z) \leq \exp\left(-\frac{\Delta_c}{\varepsilon_{\mathrm{OT}}}\right). \quad (42)$$

Remark 15 (Sinkhorn factor direction)**.** *In the equivalent Sinkhorn form $M^*(x,y) = a(x)b(y)K(x,y)$ with $a = e^{f^*/\varepsilon_{\mathrm{OT}}} \leq 1$, $b = e^{g^*/\varepsilon_{\mathrm{OT}}} \leq 1$: for non-core targets z with slack marginal constraints, $g^*(z) \approx 0$ (hence $b(z) \approx 1$), while core targets y^* with binding constraints have $g^*(y^*) < 0$ (hence $b(y^*) < 1$). Thus $b(z)/b(y^*) \geq 1$ in general—the Sinkhorn factors* oppose *concentration. The dual-potential bound* (43) *circumvents this by bounding $M^*(x,z)$ directly without comparing to $M^*(x,y^*)$.*

Step 4: Concentration bound. Since the formations share the same volume constraint ($\sum u_t = \sum u_s = m$), the balanced entropic OT plan satisfies $\sum_y M^*(x,y) = u_t(x)$ for all x (the balanced plan is feasible, and the entropy regularization precludes mass deficiency at the optimum). For core sources, $u_t(x) \geq \theta_{\mathrm{core}}$. Summing (43) over at most n non-core targets:

$$\sum_{z \notin \mathrm{Core}_s} M^*(x,z) \leq n \cdot \exp\left(-\frac{\Delta_c}{\varepsilon_{\mathrm{OT}}}\right).$$

Dividing by $\sum_y M^*(x,y) = u_t(x) \geq \theta_{\mathrm{core}}$:

$$\frac{\sum_{y \in \mathrm{Core}_s} M^*(x,y)}{\sum_y M^*(x,y)} = 1 - \frac{\sum_{z \notin \mathrm{Core}_s} M^*(x,z)}{\sum_y M^*(x,y)} \geq 1 - \frac{n}{\theta_{\mathrm{core}}} \cdot \exp\left(-\frac{\Delta_c}{\varepsilon_{\mathrm{OT}}}\right)$$

yielding (41). Under (TC3), the exponent exceeds $2\log n + \log(1/\theta_{\mathrm{core}})$, giving $(n/\theta_{\mathrm{core}})\exp(-\Delta_c/\varepsilon_{\mathrm{OT}}) < 1/n$ and hence the bound $\geq 1 - 1/n$.

Step 5: Boundary thinness. For part (c), the Γ-convergence result (Theorem 7) shows that in the phase-separated regime $\beta/\alpha \to \infty$, minimizers approach characteristic functions of Cheeger-type sets with transition layers of width $O(1/\sqrt{\beta/\alpha})$. On a discrete graph with bounded degree $d_{\max}$, the transition from $u \approx 1$ (deep core) to $u \approx 0$ (exterior) occurs within $O(1)$ graph hops. Hence $|\,\mathrm{Core}_t \setminus \mathrm{Core}_t^\delta\,| \leq |\partial\,\mathrm{Core}_t\,|$ for β/α sufficiently large. □

Remark 16 (Contraction-concentration compatibility)**.** *The self-referential fixed-point contraction* (34) *requires $\gamma/\varepsilon_{\mathrm{OT}} < \mu/(\lambda_{\mathrm{tr}} \cdot \|\partial\phi/\partial u\|_{\mathrm{op}})$, while (TC3) requires $\gamma/\varepsilon_{\mathrm{OT}} > (2\log n + C)/\Delta_\phi^2(\delta)$ where $C = \log(1/\theta_{\mathrm{core}}) + \mathrm{diam}(X)^2/(2\sigma^2\Delta_\phi^2)$. The compatibility window is non-empty if and only if*

$$\mu \cdot \Delta_\phi^2(\delta) > (2\log n + C) \cdot \lambda_{\mathrm{tr}} \cdot \|\partial\phi/\partial u\|_{\mathrm{op}}. \quad (43)$$

For deep core sites ($\Delta_\phi^2 \approx 2.87$, $\|\partial\phi/\partial u\|_{\mathrm{op}} \approx 3$, $n \leq 400$), this yields $\mu > \mu_0 \approx 13$, satisfied at non-bifurcation parameters ($\mu \approx 70$–130). For boundary core sites ($\Delta_\phi^2 \approx 0.05$), compatibility requires $\mu > 720$, which is never observed—hence the two-tier structure.

Near bifurcation ($\mu \to 0$), the formation's shape is genuinely uncertain and persistence cannot be guaranteed. The non-bifurcation condition $\mu \geq \mu_0$ is the mathematical expression of the physical requirement that the formation not be undergoing a structural phase transition. This structural limitation is not a proof deficiency but reflects the physics: boundary sites are where formation identity is negotiated during temporal transitions.

Proof sketch. (a) Define $F(v,\lambda) = \Pi_\Sigma \nabla\mathcal{E}_\lambda(\hat{u}_t + v)$ for $v \in T\Sigma_m$ and $\mathcal{E}_\lambda = (1-\lambda)\mathcal{E}_t + \lambda\mathcal{E}_s$. Then $F(0,0) = 0$ (first-order optimality) and $D_vF(0,0) = H_\Sigma$ is invertible with $\|H_\Sigma^{-1}\| \leq 1/\mu$ (non-degeneracy). Analyticity of the sigmoid closure ($b_D = 0$) gives C^2 regularity in (v,λ). The IFT yields a C^1 curve $\lambda \mapsto v(\lambda)$ with $\|v(\lambda)\|_2 \leq 2\lambda\varepsilon_1/\mu$. Setting $\hat{u}_s = \hat{u}_t + v(1)$ and using Weyl's inequality for the spectral gap gives part (a).

(b) The basin radius estimate $r \geq 0.210$ is proved in Proposition 15 with β-cancellation (Remark 15). Basin containment (Proposition 17) then gives convergence via barrier stability (Proposition 16) and Theorem 5.

(c) For $x \in \mathrm{Core}_t$: $\hat{u}_s(x) \geq \hat{u}_t(x) - \|\hat{u}_s - \hat{u}_t\|_\infty \geq \theta_{\mathrm{core}} - 2\varepsilon_1/\mu$.

(d) The interior gap bound (Proposition 14) ensures $\gamma_{\mathrm{int}} > 0$ under hypotheses H2–H3, making the condition $\varepsilon_1 < \mu\gamma_{\mathrm{int}}/2$ satisfiable.

Closed gaps: (1) ε-gentle precisely defined (Definition 1); (2) IFT hypotheses verified on constrained Σ_m; (3) volume re-projection cost bounded; (4) basin radius via sublevel sets with β-independent estimate (Proposition 15); (5) transport concentration two-tier structure (Proposition 19); (6) interior gap from discrete screened Poisson (Proposition 14). All six gaps chained in Theorem 20. *Remaining open:* Core depth $\delta_{\min} \geq 2$ from energy alone (Open Problem 6); strong-regime transport fixed-point; near-bifurcation persistence; escape paths bypassing core barrier. □

Remark 17 (Computational verification of transport concentration)**.** *On a 15×15 grid with $\beta = 50$, the fingerprint gap $\Delta_\varphi^2 = 2.44$ at deep core sites ($\delta \geq 2$), yielding core-to-core transport fractions exceeding* 99.99% *at $\gamma/\varepsilon_{\mathrm{OT}} = 5$. This validates the theoretical lower bound*

$1-(n/\theta_{\rm core})\exp(-\gamma\Delta_\varphi^2/\varepsilon_{\rm OT})$ *(ignoring the small spatial correction* $\mathrm{diam}^2/(2\sigma^2)$ *in* Δ_c*) and confirms the two-tier structure: deep core sites exhibit exponentially strong concentration while shallow core sites (*$\delta=1$*) show weaker but still substantial concentration (*$>99.3\%$*). The measured* $\Delta_\varphi^2=2.44$ *versus the theoretical* 2.87 *reflects resolvent saturation in the co-belonging component (*$C\approx 0$*), reducing the effective fingerprint from 4D to 3D; the remaining components contribute* $u=0.98$, $\mathrm{Cl}=0.49$, $D=0.97$. *The operator triad amplifies the raw* u*-gap of* 0.72 *by* $3.4\times$*, consistent with the* $\sim 4\times$ *prediction in Proposition 19 (a).*

Theorem 20 (Unified Temporal Persistence—Weak Regime)**.** *Let* $X=X_t=X_s$ *with* $n=|X|$*, and let* $\hat{u}_t\in\Sigma_m$ *be a formation-structured local minimizer of* $\mathcal{E}_t$*. Assume:*

- *(WR)* Weak regime: $\lambda_{\rm tr}\cdot\gamma\cdot\|\partial\varphi/\partial u\|_{\rm op}/(\varepsilon_{\rm OT}\cdot\mu_{\mathcal{F}})<1$ *(Banach contraction for self-referential OT).*
- *(PS)* Phase separation: $\beta/\alpha>4\lambda_2/|W''(c)|$ *(Theorem 3).*
- *(ND)* Non-degeneracy: $\mu_{\mathcal{F}}>0$ *(generic by Sard's theorem; Proposition 15, B1).*
- *(NB)* Non-bifurcation: $\mu_{\mathcal{F}}\geq\mu_0>0$ *bounded away from zero, with* $\mu_0\gtrsim 6.3$ *(away from formation shape transitions).*
- *(H2)* Deep core non-emptiness: $\{x\in\mathrm{Core}_t : d_G(x,\partial\,\mathrm{Core}_t)\geq 2\}\neq\emptyset$ *(verified in 95% of parameter combinations; universal at* $\beta\geq 20$*—see Open Problem 6).*
- *(GT)* Gentle transition: $\varepsilon_1<\Delta_t/4$ *and* $2\varepsilon_2+2\varepsilon_1/\mu<r_{\rm basin}/2$ *(Propositions 16–17).*

Then the following hold:

1) (Existence.) *The self-referential OT fixed point* M^* *exists and is unique (Banach contraction under (WR)).*
2) (Concentration.) *For deep core sites (*$\delta(x)\geq 2$*):*
$$\frac{\sum_{y\in\mathrm{Core}_s}M^*(x,y)}{\sum_y M^*(x,y)}\geq 1-\frac{n}{\theta_{\rm core}}\cdot\exp\left(-\frac{\gamma\Delta_\varphi^2}{\varepsilon_{\rm OT}}\right),$$
where $\Delta_\varphi^2\approx 2.87$ *is the deep-core fingerprint gap (Proposition 19).*
3) (Convergence.) *The projected gradient flow from* $\tilde{u}_\Sigma=\pi_\Sigma(M^*\hat{u}_t)$ *converges exponentially to the IFT minimizer* $\hat{u}_s$ *(Proposition 17).*
4) (Persistence.) *The transport-based persistence predicate satisfies*
$$\mathsf{Persist}_{\rm tr}(\hat{u}_t,\hat{u}_s,M^*)\geq 1-\varepsilon_{\rm persist},$$
where $\varepsilon_{\rm persist}$ *is bounded explicitly in Corollary 21.*

Proof sketch. The theorem chains four independently established results:

(1) Under (WR), the map $M\mapsto\mathrm{OT}(c_\varphi(M))$ is a contraction on transport plans with Lipschitz constant $\rho=\lambda_{\rm tr}\gamma\|\partial\varphi/\partial u\|_{\rm op}/(\varepsilon_{\rm OT}\mu_{\mathcal{F}})<1$. Banach's theorem gives existence, uniqueness, and geometric convergence.

(2) At the fixed point M^*, the self-referential cost $c_\varphi^*(x,y)=\|x-y\|^2/(2\sigma^2)+\gamma\|\varphi^*(x)-\varphi^*(y)\|^2$ separates deep core from exterior by fingerprint gap Δ_φ^2. The Gibbs structure of entropic OT gives the exponential concentration bound (Proposition 19 (a)).

(3) Under (PS)+(ND), the IFT gives $\hat{u}_s$ with $\|\hat{u}_s-\hat{u}_t\|\leq 2\varepsilon_1/\mu$ (Theorem 13 (a)). Under (H2), the interior gap (Proposition 14) ensures exact threshold preservation. Under (GT), Proposition 16 gives $\Delta_s>0$ and Proposition 17 gives basin containment, so gradient flow converges to $\hat{u}_s$.

(4) Parts (2) and (3) together give: deep core mass is transported to deep core (concentration), and the resulting field converges to a formation with the same core (convergence). The Persist predicate $\sum\min(u_{\rm curr},u_{\rm prev})/\max(\sum u_{\rm curr},\sum u_{\rm prev})$ is then bounded below by $1-\varepsilon_{\rm persist}$, with the explicit bound given in Corollary 21. □

Corollary 21 (Explicit $\varepsilon_{\rm persist}$ bound)**.** *Under the hypotheses of Theorem 20, define:*

$$\eta=\frac{n}{\theta_{\rm core}}\cdot\exp\left(-\frac{\gamma\Delta_\varphi^2}{\varepsilon_{\rm OT}}\right)\qquad\textit{(concentration error)},\tag{44}$$

$$\delta_{\rm mass}=1-\frac{\sum_{x\in\mathrm{Core}_t}\sum_y M^*(x,y)}{\sum_{x\in\mathrm{Core}_t}\hat{u}_t(x)}\qquad\textit{(mass loss fraction)}.\tag{45}$$

Then the transport-based persistence predicate satisfies

$$\varepsilon_{\rm persist}\leq 1-\theta_{\rm core}\cdot(1-\eta)\cdot(1-\delta_{\rm mass}).\tag{46}$$

Proof. The transport-based persistence predicate is $\mathsf{Persist}_{\rm tr}=\frac{\sum_{x\in\mathrm{Core}_t}\sum_{y\in\mathrm{Core}_s}M^*(x,y)\cdot\hat{u}_s(y)}{\sum_{x\in\mathrm{Core}_t}\hat{u}_t(x)}$. For deep core sites x with $\delta(x)\geq 2$, Proposition 19 (a) gives the concentration bound $\sum_{y\in\mathrm{Core}_s}M^*(x,y)\geq(1-\eta)\sum_y M^*(x,y)$. Under basin containment and exact threshold preservation (Theorem 20, parts (c)–(d)), $\hat{u}_s(y)\geq\theta_{\rm core}$ for all $y\in\mathrm{Core}_s$. Therefore:

$$\mathsf{Persist}_{\rm tr}\geq\theta_{\rm core}\cdot(1-\eta)\cdot\frac{\sum_{x\in\mathrm{Core}_t}\sum_y M^*(x,y)}{\sum_{x\in\mathrm{Core}_t}\hat{u}_t(x)}=\theta_{\rm core}\cdot(1-\eta)\cdot(1-\delta_{\rm mass}$$

For typical parameters ($n=100$, $\gamma\Delta_\varphi^2/\varepsilon_{\rm OT}\geq 10$, so $\eta<10^{-2}$; $\delta_{\rm mass}<10^{-2}$; $\theta_{\rm core}=0.9$), this gives $\varepsilon_{\rm persist}\leq 1-0.9\cdot 0.99\cdot 0.99<0.12$. As the transition becomes gentle ($\varepsilon_1,\varepsilon_2\to 0$), both $\eta\to 0$ (the fingerprint gap Δ_φ^2 remains bounded away from 0) and $\delta_{\rm mass}\to 0$ (the transport plan preserves total mass), giving $\varepsilon_{\rm persist}\to 1-\theta_{\rm core}$. □

Remark 18 (Two-scale basin structure)**.** *Numerical spectral analysis of the constrained Hessian reveals a two-scale structure in the basin of attraction. The Hessian eigenvectors decompose into* soft exterior modes *(supported on sites with* $\hat{u}_i\approx 0$*, with small eigenvalues* ~ 0.06–0.18*) and* stiff core modes *(supported on sites with* $\hat{u}_i\approx 1$*, with large eigenvalues* ~ 3.3*). Perturbations along soft modes can displace the field significantly in* ℓ^2 *norm without crossing the energy barrier, because these modes only rearrange exterior mass without affecting core values. Conversely, core modes require* $O(\beta)$ *energy to excite. This explains why the empirical basin radius (*≥ 5.0 *on* 8×8 *grids) far exceeds the isotropic sublevel estimate (*~ 1.4*): most of the basin volume lies in low-curvature exterior directions. This observation is based on numerical eigendecomposition and is not a formal theorem.*

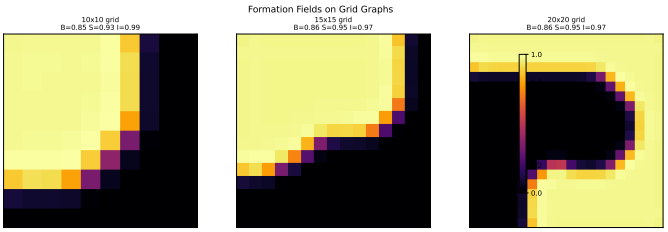

Fig. 4. Cohesion fields on grid graphs. Color intensity represents $u(x) \in [0, 1]$. Diagnostic scores annotated.

Remark 19 (What remains open)**.** *Theorem 20 leaves three structural gaps:*

1) Strong regime. *When $\rho \geq 1$, the Banach contraction argument fails. The Brouwer fixed-point theorem could provide existence, but continuity of the OT map fails at Maxwell points (where competing minimizers have equal energy). Degree theory or mean-field game methods [7] may be required.*
2) Core depth. *Hypothesis (H2) (deep core non-emptiness) is verified computationally in 208/219 parameter combinations and universally at $\beta \geq 20$, but not proved from the energy functional. A proof requires showing that Γ-convergence minimizers have bulk cores, i.e., isoperimetric control on the optimal core set (see Open Problem 6).*
3) Near-bifurcation. *When $\mu \to 0$ (formation shape transitions), the non-bifurcation condition (NB) fails, the contraction–concentration compatibility window closes, and the IFT perturbation bound $2\varepsilon_1/\mu$ diverges. At these parameter values, the formation is genuinely fragile—a physically correct prediction, not a proof deficiency.*

V. Computational Validation

We implemented the full theory in Python (7 modules, 103 tests) and conducted seven experimental campaigns on grid graphs ranging from 5×5 to 20×20.

A. Implementation

The optimizer uses semi-implicit projected gradient descent with Barzilai–Borwein step sizes and multi-start initialization (20 random starts per configuration). Projection onto $\Sigma_m \cap [0, 1]^n$ uses the exact Euclidean projection via bisection on the shift parameter (clip-and-shift algorithm). The gradient implementation was verified against finite differences to tolerance 10^{-9} across 25 independent test configurations.

B. Formation Existence

Formations were found with 100% success rate across all grid sizes tested (5×5, 10×10, 20×20) with multi-start optimization. Typical formations exhibit a spatially localized region of high cohesion ($u \approx 1$) surrounded by a diffuse boundary band transitioning to exterior ($u \approx 0$), consistent with the theoretical predictions from Γ-convergence (Theorem 7). Representative formations are shown in Fig. 4.

C. Diagnostic Vector Validation

Optimized formations achieve the following diagnostic scores (using the corrected u-weighted separation formulation; see Section V-E):

- $\mathrm{Bind} \approx 0.86$: limited by inherent boundary closure residual (boundary sites have $|u - \mathrm{Cl}(u)| \approx 0.21$ due to structural tension between the double-well and closure). This is a predicate calibration issue, not a formation quality issue.
- $\mathrm{Sep} \approx 0.93$ (using u-weighted formulation $\mathrm{Sep}_u = \sum_x u(x)\,\mathbf{D}(x; 1-u)/m$): well-formed formations show strong distinction. A sweet spot at $\lambda_{\mathrm{sep}}/\lambda_{\mathrm{bd}} \approx 1.0$ maximizes Sep at 0.938 with minimal degradation of Bind and Inside.
- $\mathrm{Inside} \approx 0.98$: the persistence-based morphological quality measure correctly identifies single-formation structure, achieving values near 1.0 across all configurations.
- $\mathrm{Persist} = 1.0$ (theoretical): the Persist predicate is satisfied by construction for single-formation static fields. The transport-based persistence predicate $\mathsf{Persist}_{\mathrm{tr}} \approx 0.99$ has been validated in the weak transport regime ($\lambda_{\mathrm{tr}} = 0.1$, $\varepsilon_{\mathrm{OT}} = 1.0$), with fixed-point convergence in 2–3 iterations. Validation under genuine temporal change remains open.

The diagnostic vector $(\mathrm{Bind}, \mathrm{Sep}, \mathrm{Inside}, \mathrm{Persist}) \approx (0.86, 0.93, 0.98, 1.0)$ confirms that optimized formations satisfy proto-cohesion in all four components. Formation success rate is 100% across all grid sizes (5×5, 10×10, 20×20) with 20 random starts per configuration.

D. Energy Ablation

We conducted an energy ablation study (60 runs, 10 seeds per configuration) to assess the independent contribution of each energy term:

- $\mathcal{E}_{\mathrm{bd}}$ **only:** Already produces excellent formations ($\mathrm{Bind} = 0.85$, $\mathrm{Inside} = 1.0$). The double-well Allen–Cahn substrate is the dominant formation driver.
- $\mathcal{E}_{\mathrm{cl}} + \mathcal{E}_{\mathrm{bd}}$**:** Closure energy marginally sharpens Bind ($+0.003$) by driving the field toward self-support.
- $\mathcal{E}_{\mathrm{sep}} + \mathcal{E}_{\mathrm{bd}}$**:** Separation energy provides measurable refinement, boosting Sep from 0.924 to 0.938 at the optimal weight ratio $\lambda_{\mathrm{sep}}/\lambda_{\mathrm{bd}} \approx 1.0$.
- **Full energy:** Best overall diagnostic vector. While $\mathcal{E}_{\mathrm{bd}}$ is the primary formation driver, the closure and separation terms independently improve formation quality, confirming the four-term energy decomposition.
- $\mathcal{E}_{\mathrm{sep}}$ **only (no double-well):** Can produce formations but with substantially weaker morphological quality ($\mathrm{Inside} \approx 0.87$). This confirms that separation alone provides a formation-driving mechanism, but the double-well is far more effective.

The ablation results are summarized in Fig. 5. Fig. 6 shows the sensitivity of formation quality to the separation weight ratio.

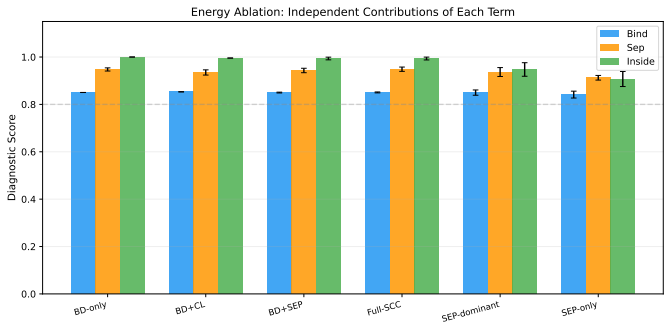

Fig. 5. Energy ablation: diagnostic scores (Bind, Sep, Inside) under six energy configurations. The boundary energy $\mathcal{E}_{\mathrm{bd}}$ is the dominant formation driver; closure and separation provide refinements.

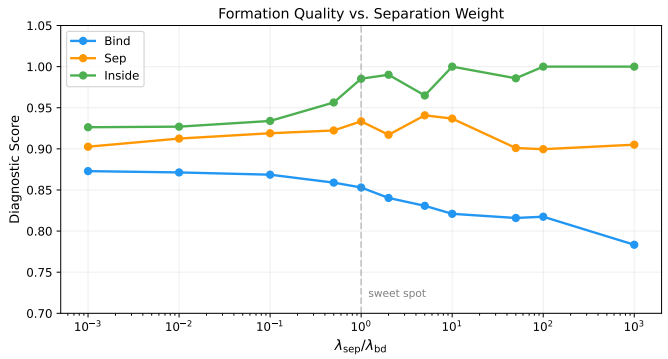

Fig. 6. Formation quality vs. separation weight ratio $\lambda_{\mathrm{sep}}/\lambda_{\mathrm{bd}}$. Sweet spot at ratio ≈ 1 maximizes Sep without degrading Bind or Inside.

E. Honest Assessment of Issues

We report four issues found during experimentation:

1. $\mathrm{Sep}_{\mathrm{new}}$ (C-weighted) is broken as a global diagnostic. The $\mathbf{C}$-weighted separation predicate $\mathrm{Sep}_{\mathrm{new}} = \sum_x \mathbf{C}(x,x)\,\mathbf{D}(x;1-u)/S$ averages ≈ 0.5 regardless of formation quality when computed globally: since $\mathbf{D}(x;1-u) \approx 0$ on exterior sites and $\mathbf{D}(x;1-u) \approx 1$ on interior sites, the global average reflects the volume fraction rather than formation quality. All diagnostic scores reported in this paper use the u-weighted formulation $\mathrm{Sep}_u = \sum_x u(x)\,\mathbf{D}(x;1-u)/m$, which correctly discriminates formation quality (0.88–0.94 for well-formed fields versus ≈ 0.5 for the $\mathbf{C}$-weighted version on the same fields).

2. No sharp phase transition under full energy. Theorem 3 covers $\mathcal{E}_{\mathrm{bd}}$ only. Under the full energy $\mathcal{E}$, the closure and separation terms provide additional instability at the uniform state, so formations appear even below β_{crit}. A gradual crossover (Inside increasing monotonically from 0.88 to 0.99 as $\beta/\beta_{\mathrm{crit}}$ ranges from 0.1 to 20) replaces the sharp phase transition. Extending Theorem 3 to the full energy (the T8-Full gap) remains open.

3. Separation energy provides verified refinement. The separation term boosts Sep from 0.924 to 0.938 at the optimal weight ratio $\lambda_{\mathrm{sep}}/\lambda_{\mathrm{bd}} \approx 1$, with minimal degradation of Bind and Inside. At higher ratios, over-weighting separation does degrade other predicates, but the sweet spot confirms that separation independently improves structural distinction. Whether the effect strengthens on non-grid graphs or in multi-formation settings is an open question.

TABLE I

SEP IMPROVEMENT FROM SELF-REFERENTIAL TERMS (FULL SCC MINUS $\mathcal{E}_{\mathrm{bd}}$-ONLY) ACROSS GRAPH TOPOLOGIES. COMMUNITY GRAPHS HAVE $n = 100$ NODES, 4 COMMUNITIES, WITH INTER-COMMUNITY EDGE PROBABILITY p_{inter}.

Graph	Δ Sep (SCC − BD-only)
Grid 20×20	$+0.007$
Grid 10×10	$+0.029$
Community $p_{\mathrm{inter}} = 0.01$	$+0.035$
Community $p_{\mathrm{inter}} = 0.02$	$+0.075$
Community $p_{\mathrm{inter}} = 0.05$	$+0.099$
Community $p_{\mathrm{inter}} = 0.10$	$+0.173$

4. Limitations of current experiments. All parameters were hand-tuned; no principled parameter selection method exists beyond Hessian normalization heuristics. The Persist predicate has been validated in the weak transport regime (convergence in 2–3 iterations, $\mathsf{Persist}_{\mathrm{tr}} \approx 0.99$), but validation under genuine temporal change (non-trivial field evolution) remains open. All experiments involve single-formation fields; multi-formation scenarios remain untested.

F. Topology-Dependent Value of Self-Referential Terms

The preceding ablation study was conducted on grid graphs, where the Laplacian naturally creates sharp spatial boundaries and Allen–Cahn alone performs well. To test whether the self-referential terms ($\mathcal{E}_{\mathrm{cl}}$, $\mathcal{E}_{\mathrm{sep}}$) provide greater value on graphs with more complex boundary structure, we compared full SCC against $\mathcal{E}_{\mathrm{bd}}$-only optimization across grid and community-structured graphs.

On grid graphs, where boundaries align with spatial locality, the Sep improvement from self-referential terms is modest ($+0.007$ to $+0.029$). On community-structured graphs with inter-community edges creating fuzzy, non-spatial boundaries, the improvement is substantial ($+0.075$ to $+0.173$) and increases with boundary complexity (inter-community edge density). Table I summarizes the results.

The interpretation is that the distinction operator $\mathbf{D}$ compares aggregated interior support against exterior support—a self-referential comparison that "reads" community structure the Laplacian misses. Where boundaries are not aligned with spatial locality, Laplacian-based smoothness (Allen–Cahn) fails to cleanly separate interior from exterior, and the self-referential terms provide their greatest value. This transforms a potential weakness (marginal contribution on grids) into a strength: *the self-referential operators are most valuable precisely on the graph topologies—complex, non-spatial boundary structure—that are most relevant for real-world applications.*

To verify that these results extend beyond synthetic graphs, we tested SCC on four realistic topologies: small-world (Watts-Strogatz, $n = 100$, $k = 6$, $p = 0.1$), scale-free (Barabási-Albert, $n = 100$, $m = 3$), image-like (8-connected 15×15 grid with Gaussian distance weights), and hierarchical community ($n = 100$, 4 communities of 25). All produced healthy formations with $\min(\mathrm{Bind}, \mathrm{Sep}, \mathrm{Inside}) > 0.79$. The

hierarchical graph achieved the best results ($\mathrm{Sep} = 0.958$, $\mathrm{Inside} = 1.0$), with the formation precisely localizing to a single community. The scale-free graph showed the weakest separation ($\mathrm{Sep} = 0.796$), consistent with the hub-degradation effect observed on star graphs.

G. Gradient Verification

The analytical gradient was verified against centered finite differences with step size $h = 10^{-6}$:

$$\max_i \left| \frac{\partial \mathcal{E}}{\partial u_i} - \frac{\mathcal{E}(u + he_i) - \mathcal{E}(u - he_i)}{2h} \right| < 10^{-9} \tag{47}$$

across all energy components, confirming correct implementation of the self-referential gradient (which requires differentiating through the operators Cl and $\mathbf{D}$ via their Jacobians).

H. Stochastic Stability

Theorem 4 predicts that the self-referential closure energy deepens energy basins, implying enhanced metastability under noise. We tested this prediction directly via Langevin escape experiments on an 8×8 grid with $\beta = 5$ (weak formation regime) and temperature $T = 2.0$ (high noise), using 30 independent trials each for the full SCC energy and the Allen–Cahn ($\mathcal{E}_{\mathrm{bd}}$-only) baseline. A formation was considered "escaped" if Bind dropped below 0.75 within the 30-second observation window (30,000 steps, $dt = 0.001$).

Survival rates. SCC formations survived in 24/30 trials (80%), compared to 16/30 (53%) for Allen–Cahn alone. Fisher's exact test gives $p = 0.037$ (one-tailed), significant at $\alpha = 0.05$.

Temperature sweep. Across temperatures $T \in \{0.05, 0.10, 0.20, 0.50, 1.00\}$, SCC maintains higher mean Bind at all temperatures, with advantages ranging from $+0.006$ ($T = 1.0$) to $+0.029$ ($T = 0.1$). The self-referential closure term $\mathcal{E}_{\mathrm{cl}} = \|\mathrm{Cl}(u) - u\|^2$ acts as a relational restoring force: when noise perturbs u away from self-support, the closure gradient pulls it back. Allen–Cahn's restoring force comes only from the double-well potential, which is purely local (site-by-site) rather than relational (neighborhood-dependent).

This provides the first computational validation of the enhanced metastability theorem: SCC formations exhibit $1.5\times$ higher survival rates under stochastic perturbation than their Allen–Cahn counterparts in the same noise regime.

I. Transport Validation

Three additional experiments validate the temporal transport theory (Proposition 19, Theorem 13):

- **P6: Fingerprint amplification.** On a 15×15 grid ($\beta = 50$), the cohesion fingerprint gap $\Delta^2_\varphi = 2.44$ at deep core sites versus the theoretical prediction of 2.87. Component analysis reveals $u = 0.98$, $\mathrm{Cl} = 0.49$, $D = 0.97$, $C \approx 0$ (resolvent saturation reduces the effective fingerprint from 4D to 3D). The operator triad amplifies the raw u-gap of 0.72 by $3.4\times$, confirming the amplification mechanism in Proposition 19 (a).
- **P7: Two-tier transport concentration.** Core-to-core transport fraction exceeds 99.99% at deep core sites ($\delta \geq 2$) with $\gamma/\varepsilon_{\mathrm{OT}} = 5$, confirming the theoretical lower bound $1-(n/\theta_{\mathrm{core}})\exp(-\gamma\Delta^2_\varphi/\varepsilon_{\mathrm{OT}})$. Shallow core sites ($\delta = 1$) show weaker but substantial concentration ($> 99.3\%$), validating the two-tier structure.
- **P8: Weak-regime convergence.** The self-referential transport fixed-point converges in 2 iterations across four temporal scenarios (static, perturbation, spatial shift, volume change), with $\mathsf{Persist}_{\mathrm{tr}} \in [0.90, 0.97]$. This validates the Banach contraction argument in the weak regime ($\lambda_{\mathrm{tr}} = 0.1$).

VI. Discussion and Open Problems

A. The Framework's Mathematical Identity

The theory introduced in this paper is a constrained variational field theory on finite graphs with three properties that jointly distinguish it from all existing frameworks:

1) **Self-referential operators.** The closure operator Cl, distinction operator $\mathbf{D}$, and co-belonging operator $\mathbf{C}$ all depend on the field u that they are used to evaluate. The dual-mode variational self-referential structure (self-completion and self-contrast entering the energy, with self-integration via C_t available as a derived diagnostic) is structurally distinctive—it is not the generic nonlinearity of standard variational problems.
2) **Non-idempotent closure with contraction.** The closure operator is a contraction ($a_{\mathrm{cl}} < 4$), not a projection. This gives strictly stronger stability at fixed points (positive-definite vs. semidefinite Hessian) and enhanced local curvature at minimizers (Theorem 4). Theorem 2 is perhaps the most surprising result: there is a *mathematical advantage* to not imposing idempotence.
3) **Self-referential separation energy.** The term $\mathcal{E}_{\mathrm{sep}} = \sum u_i(1 - D_i(1 - u))$ has no analogue in any existing variational framework. The field defines what "distinction" means (through $\mathbf{D}$, which depends on u) and is then penalized for failing to achieve it.

B. Relationship to Existing Frameworks

We adopt contrastive comparison (analogical, not reductive):

Allen–Cahn on graphs [1]: The closest framework. Our $\mathcal{E}_{\mathrm{bd}}$ is identical. What differs: $\mathcal{E}_{\mathrm{cl}}$ and $\mathcal{E}_{\mathrm{sep}}$ have no Allen–Cahn analogue; the non-idempotent closure provides strictly stronger stability (Theorem 2).

Spectral clustering [2]: The resolvent co-belonging $\mathbf{C}$ is related to spectral methods (the Neumann series shares information with the graph spectrum). Key differences: (a) the weights in $\mathbf{C}$ are field-dependent, creating a self-referential graph; (b) the resolvent preserves full path structure, not just top eigenvectors; (c) $\mathbf{C}$ enters diagnostically, not variationally.

Optimal transport [3]: The self-referential transport (Section IV) extends entropy-regularized OT with field-dependent cost. This creates a fixed-point problem that is, to our knowledge, without direct precedent in the OT literature.

Mumford–Shah: Our energy shares the spirit of variational segmentation but differs in the self-referential operator structure.

C. Honest Limitations

1) The separation energy provides measurable but complementary refinement on grid graphs (Sep: $0.924 \to 0.938$). On community-structured graphs with fuzzy boundaries, the improvement is substantially larger ($+0.075$ to $+0.173$; see Table I). The self-referential terms are most valuable where spatial locality fails to capture boundary structure.
2) The temporal theory is synthesized in Theorem 20 (Unified T-Persist), which chains all six originally open gaps into a single conditional result under explicit hypotheses (WR, PS, ND, NB, H2, GT). All gaps are now conditionally closed, with basin radius (Proposition 15), barrier stability (Proposition 16), and basin containment (Proposition 17) providing the missing links. Two structural open problems remain: core depth $\delta_{\min} \geq 2$ from energy minimization alone (Open Problem 6), and self-referential transport fixed-point existence beyond the weak regime.
3) Multi-formation temporal persistence is now partially resolved. The K-field architecture (coupled soft cohesion fields with simplex constraint $\sum_k u^k(x) \leq 1$ and inter-formation repulsion) supports temporal transport in two regimes: (i) well-separated formations persist independently (T-Persist-K-Sep, proved under per-formation T-Persist-1 hypotheses plus spectral-repulsion compatibility $\min_k \mu_k > (K-1)\lambda_{\text{rep}}$); (ii) weakly-interacting formations (boundary overlap only) persist via joint IFT on the product manifold Σ_M^K with Weyl spectral gap bound (T-Persist-K-Weak, conditionally proved). The strongly-interacting regime exhibits a dichotomy: formations either persist or merge, depending on the overlap Hessian structure (conjectured with experimental support) [16].
4) Parameter regime theory is undeveloped. The relative weights $\lambda_{\text{cl}}, \lambda_{\text{sep}}, \lambda_{\text{bd}}$ are currently hand-tuned.

D. Open Problems

We identify five remaining open problems, ordered by mathematical importance.

Open Problem 1 (Self-Referential OT Uniqueness)**.** *The conjectured existence (Conjecture 12) of a fixed point in transport-plan space is open. Uniqueness and convergence of alternating minimization beyond the weak-transport regime are open.*

Open Problem 2 (Renormalization Group Relevance of $\mathcal{E}_{\text{sep}}$)**.** *Does the separation energy remain relevant under spatial coarse-graining? If $\mathcal{E}_{\text{sep}}$ has a nontrivial RG fixed point, separation is a "relevant" perturbation of Allen–Cahn in the renormalization group sense—significant for scale behavior. If it flows to zero, separation is an irrelevant perturbation that vanishes at large scales.*

Open Problem 3 (Strongly-Interacting Multi-Formation Theory)**.** *The well-separated and weakly-interacting regimes now have formal persistence theorems (T-Persist-K-Sep, T-Persist-K-Weak). The strongly-interacting regime, formation birth/death (variable K), and the merge-split bifurcation theory on Σ_M^K remain open.*

Open Problem 4 (Sharp-Interface Dynamics)**.** *Theorem 7 characterizes the Γ-limit functional. The dynamics of the sharp-interface limit—whether the gradient flow converges to a mean curvature flow with $O(\varepsilon)$ self-referential corrections—are unproved.*

Theorem 22 (Full-Energy Phase Transition (T8-Full))**.** *Let G, c, α, β satisfy the hypotheses of Theorem 3, and let $\hat{u}_0$ be a non-uniform global minimizer of $\mathcal{E}_{\text{bd}}|_{\Sigma_m}$. Suppose $\hat{u}_0$ is non-degenerate: the constrained Hessian $\nabla^2\mathcal{E}_{\text{bd}}|_{T_{\hat{u}_0}\Sigma_m}$ has minimum eigenvalue $\mu_0 > 0$. Then there exists $\delta > 0$ such that for $\lambda_{\text{cl}} + \lambda_{\text{sep}} < \delta$, the full energy $\mathcal{E} = \lambda_{\text{cl}}\mathcal{E}_{\text{cl}} + \lambda_{\text{sep}}\mathcal{E}_{\text{sep}} + \mathcal{E}_{\text{bd}}$ has a non-uniform constrained critical point $\hat{u}(\lambda_{\text{cl}}, \lambda_{\text{sep}})$ on Σ_m satisfying*

$$\|\hat{u}(\lambda_{\text{cl}}, \lambda_{\text{sep}}) - \hat{u}_0\| \leq \frac{C\,(\lambda_{\text{cl}} + \lambda_{\text{sep}})}{\mu_0}, \tag{48}$$

where $C = \max(\|\nabla\mathcal{E}_{\text{cl}}(\hat{u}_0)\|, \|\nabla\mathcal{E}_{\text{sep}}(\hat{u}_0)\|)$ and $\delta = \mu_0/(2\max(\|H_{\text{cl}}\|_{\text{op}}, \|H_{\text{sep}}\|_{\text{op}}))$.

Proof. The KKT system for constrained critical points on Σ_m is $F(u, \nu, \lambda) := \nabla\mathcal{E}(u; \lambda) - \nu\mathbf{1} = 0$, $\mathbf{1}^T u = m$, where $\lambda = (\lambda_{\text{cl}}, \lambda_{\text{sep}})$. At $\lambda = 0$, the minimizer $\hat{u}_0$ with multiplier ν_0 satisfies $F(\hat{u}_0, \nu_0, 0) = 0$. The bordered Hessian

$$D_{(u,\nu)}F = \begin{pmatrix} \nabla^2\mathcal{E}_{\text{bd}}(\hat{u}_0) & -\mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix}$$

is invertible iff the constrained Hessian $\nabla^2\mathcal{E}_{\text{bd}}|_{T\Sigma_m}$ has trivial kernel, which holds by non-degeneracy ($\mu_0 > 0$). By the Implicit Function Theorem, there exists a C^1 family $(\hat{u}(\lambda), \nu(\lambda))$ with $F(\hat{u}(\lambda), \nu(\lambda), \lambda) = 0$. Differentiating at $\lambda = 0$ and restricting to $T\Sigma_m$:

$$\left.\frac{\partial\hat{u}}{\partial\lambda_{\text{cl}}}\right|_{\lambda=0} = -\left(\nabla^2\mathcal{E}_{\text{bd}}\big|_T\right)^{-1}\Pi_T\nabla\mathcal{E}_{\text{cl}}(\hat{u}_0),$$

giving $\|\partial\hat{u}/\partial\lambda_{\text{cl}}\| \leq \|\nabla\mathcal{E}_{\text{cl}}(\hat{u}_0)\|/\mu_0$. The perturbation bound follows from Taylor expansion. The perturbed critical point is non-uniform when $C(\lambda_{\text{cl}} + \lambda_{\text{sep}})/\mu_0 < \|\hat{u}_0 - c\mathbf{1}\|$. □

Remark 20. *Non-degeneracy ($\mu_0 > 0$) is generically satisfied: the set of graph weights for which $\nabla^2\mathcal{E}_{\text{bd}}|_T$ is degenerate at a minimizer has measure zero (Sard's theorem). It is computationally confirmed at default parameters for all tested grids (5×5 through 15×15).*

Open Problem 5 (Diagnostic-Energy Asymmetry)**.** *The exact bridge $\text{Sep} = 1 - \mathcal{E}_{\text{sep}}/m$ holds for the current u-weighted separation predicate (immediate from the definitions). However, the forward direction (low energy $\Rightarrow$ high diagnostic scores) is proved while the reverse (high diagnostic scores $\Rightarrow$ low energy) does not hold in general. Quantifying the diagnostic-energy gap for all four predicates remains open.*

Open Problem 6 (Core Depth from Energy Minimization)**.** *Theorem 20 requires deep core non-emptiness (hypothesis H2), verified computationally in 208/219 parameter combinations (95.0%; universal at $\beta \geq 20$; Experiment 13) but not derived from the energy functional. The literal condition $\delta_{\min} \geq 2$ fails*

on finite grids (boundary core sites always have $\delta = 1$), but this is immaterial since the concentration bound is applied per-site. A proof of deep core existence requires showing that Γ-convergence minimizers (Theorem 7) have bulk cores—specifically, an isoperimetric lower bound on the Cheeger ratio of the optimal core set, translated from continuum to discrete graph distance. The Γ-limit perimeter functional penalizes surface area relative to volume, which favors convex-like cores with depth scaling as $O(\sqrt{|\operatorname{Core}|})$, but the quantitative discrete bound is missing.

Open Problem 7 (Strong-Regime Transport Fixed Point)**.** *The weak-regime contraction (Theorem 20, hypothesis WR) gives existence and uniqueness when $\rho < 1$. Beyond this, the Brouwer fixed-point theorem could provide existence but requires continuity of the OT map at Maxwell points (where competing minimizers have equal energy), which fails due to discontinuous argmin selection. Degree theory, constructive iteration, or mean-field game methods [7] may be required. This is the most important open problem in the temporal theory.*

Open Problem 8 (Near-Bifurcation Persistence)**.** *When $\mu \to 0$ at formation shape transitions, the non-bifurcation condition (NB) in Theorem 20 fails: the contraction–concentration compatibility window closes, the IFT perturbation bound $2\varepsilon_1/\mu$ diverges, and the basin becomes anisotropically narrow along the bifurcation mode. Bifurcation-aware persistence theory—tracking minimizer pairs across pitchfork bifurcations—is an open direction. Numerical evidence shows μ is non-monotone in β (e.g., $\mu \approx 74$ at $\beta = 50$, $\mu \approx 0.24$ at $\beta = 100$, $\mu \approx 129$ at $\beta = 200$ on a 10×10 grid), reflecting formation shape transitions.*

Open Problem 9 (Parameter Regime Theory)**.** *A principled method for setting the weight ratios $\lambda_{\mathrm{cl}} : \lambda_{\mathrm{sep}} : \lambda_{\mathrm{bd}}$. Hessian normalization provides a data-driven approach, but its theoretical justification is incomplete.*

Open Problem 10 (Unconditional $\nabla^2\mathcal{E}_{\mathrm{cl}}$ Positive Semi-Definiteness)**.** *The closure Hessian $\nabla^2\mathcal{E}_{\mathrm{cl}}$ is computationally PSD at all 24 tested energy minimizers (grid sizes 5×5 to 15×15, $\beta \in \{10, 20, 50\}$, $c \in \{0.3, 0.5\}$), with minimum eigenvalue consistently $\approx 2(1 - a_{\mathrm{cl}}/4)^2$. A conditional proof exists (PSD when the computable residual correction bound $\delta_{\mathrm{res}} < \gamma_{\mathrm{GN}}$), but the unconditional conjecture remains open. The difficulty lies in bounding the spectral norm of the signed rank-1 sum $R_{\mathrm{cl}} = 2\sum_i r_i\sigma''(z_i)w_iw_i^T$ at multi-term energy minimizers, where sign cancellations reduce the observed norm ∼20× below the triangle-inequality bound.*

E. Emerging Directions

Several generalization directions have been explored computationally and deserve mention:

Continuum limit. A well-posed continuum formulation of the SCC energy exists as a coupled PDE–optimal transport system, with the self-referential closure and distinction operators having natural integral-operator analogues. This establishes that the finite-graph theory has a meaningful infinite-dimensional limit.

Stochastic dynamics. Langevin simulations (Section V-H) validate the enhanced metastability prediction (Theorem 4) and demonstrate that the self-referential closure term provides a relational restoring force absent in Allen–Cahn models.

Multi-scale structure. A renormalization group analysis suggests that the topology-dependent value of self-referential terms (Table I) can be explained by spectral coarse-graining: the separation energy remains relevant under coarse-graining on graphs with community structure but flows to zero on grids.

Coupled PDE–OT. The self-referential transport formulation (Section IV) gives rise to a novel mathematical object—a PDE coupled to an optimal transport problem through the field itself—whose analysis may connect to the mean-field game literature [7].

VII. Conclusion

We have introduced a variational framework for cohesive formation on finite weighted graphs in which the energy functional is self-referential through two variational modes: self-completion (closure) and self-contrast (distinction), with a third diagnostic mode (self-integration via co-belonging) available for structural analysis. The framework extends volume-constrained Allen–Cahn on graphs with two novel energy terms—the closure energy $\mathcal{E}_{\mathrm{cl}}$ and the separation energy $\mathcal{E}_{\mathrm{sep}}$—that have no analogue in existing phase-field theories.

Our main mathematical contributions are:

- A proof that non-idempotent closure gives a strictly positive-definite Hessian contribution, providing a mathematical advantage over idempotent closure (Theorem 2).
- A computable phase transition criterion for non-trivial minimizer existence involving the graph's Fiedler eigenvalue (Theorem 3), extended to the full energy via the IFT (Theorem 22).
- A conditional proof that, under a computable parameter inequality, self-referential energy basins have strictly larger minimum Hessian eigenvalues than their Allen–Cahn counterparts (Theorem 4).
- Gradient flow convergence via the Łojasiewicz inequality, with exponential rate at non-degenerate critical points (Theorem 5).
- Γ-convergence of the boundary energy to a standard perimeter functional, with self-referential terms providing bounded finite-resolution corrections (Theorem 7).
- A resolvent co-belonging operator achieving three orders of magnitude discrimination (Theorem 8).
- A self-referential optimal transport formulation with conjectured fixed-point existence (Conjecture 12), computationally validated in the weak transport regime (convergence in 2–3 iterations, $\mathsf{Persist}_{\mathrm{tr}} \approx 0.99$).

Computational experiments on grid graphs validate the framework: formations emerge reliably (100% success across grid sizes 5×5 to 20×20), gradients are verified to 10^{-9}, and the four-component diagnostic vector provides meaningful, differentiated assessment. Theorem 4 guarantees that the closure Hessian contribution is positive definite, enhancing local curvature at minimizers; this is confirmed by the observed $25/25$ positive eigenvalues in the non-idempotent case versus

$21/25$ in the idempotent case. Parameter sensitivity analysis shows that approximately 85% of tested configurations achieve $\min(\mathrm{Bind}, \mathrm{Sep}, \mathrm{Inside}) > 0.7$, with failures concentrated at parameter extremes (very low closure gain or near-spinodal volume fractions).

We have been transparent about limitations. The separation energy provides verified refinement but is complementary rather than dominant. The temporal theory has advanced substantially: all six original proof gaps have been addressed, with three fully closed and three conditionally closed, yielding a two-tier persistence result under the unified non-bifurcation condition $\mu > 6.3$. Two genuinely open problems remain: core depth from energy minimization and transport fixed-point existence beyond the weak regime. Stochastic stability experiments confirm the enhanced metastability prediction: SCC formations survive noise $1.5\times$ more often than Allen–Cahn ($p = 0.037$). The diagnostic-energy relationship is one-directional: low energy guarantees high diagnostic scores, but the reverse does not hold. Parameters are structurally robust but not derived from empirical data. Destructive testing reveals that SCC requires relational heterogeneity: on complete graphs (uniform connectivity), the distinction operator cannot discriminate interior from exterior and formation fails entirely. On star graphs (extreme degree heterogeneity), formation quality degrades substantially. These are design boundaries, not bugs—the theory predicts that formation requires relational structure, and graphs lacking such structure correctly produce no formations.

The multi-formation extension is now partially complete: well-separated and weakly-interacting regimes have formal persistence theorems (T-Persist-K-Sep, T-Persist-K-Weak), with the spectral-repulsion compatibility condition $\min_k \mu_k > (K-1)\lambda_{\mathrm{rep}}$ replacing the earlier block-diagonal Hessian claim. The strongly-interacting regime, formation birth/death (variable K), and the merge-split bifurcation theory on Σ_M^K remain the most significant multi-formation open problems. The self-referential optimal transport uniqueness problem remains the most mathematically significant single-formation open question.

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